

Multimodal Learning

Luca Gonzalez-Sommer

Artificial and Natural Intelligence Toulouse Institute (ANITI)

2nd April, 2026



I. Introduction

II. Recall

A- MLP

B- CNN

C- RNN

D- Transformer

E- How to train a model

II. How to learn from Multimodality ?

III. Multimodal Tasks

A- Image Captioning

B- VQA

C- Multimodal Dialogue

D- Language, Vision and Navigation

IV. Examples of Models

V. Conclusion

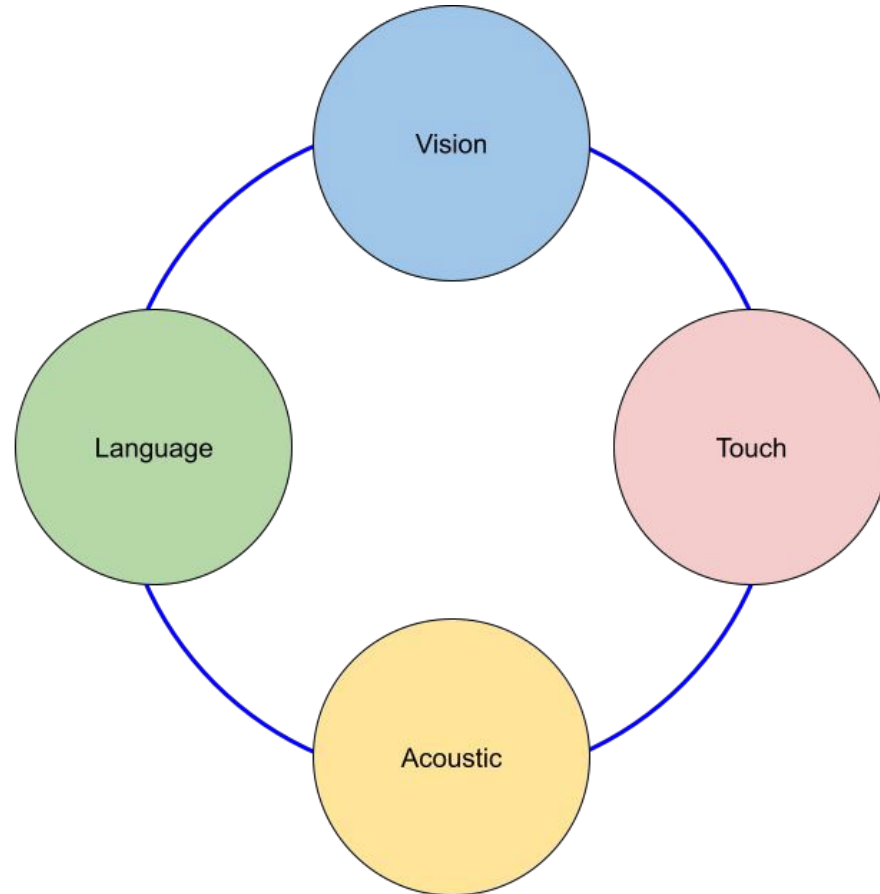
Introduction

I- Introduction

Multimodal Machine Learning is the study of computer algorithms that learn and improve through the use and experience of data from multiple modalities

Multimodal Artificial Intelligence studies computer agents able to demonstrate intelligence capabilities such as understanding, reasoning and planning, through multimodal experiences, and data

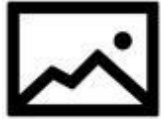
I- Introduction



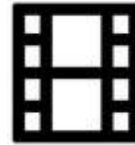
I- Introduction

Why do we need multimodal data ?

I- Introduction



Limited World Model and Knowledge



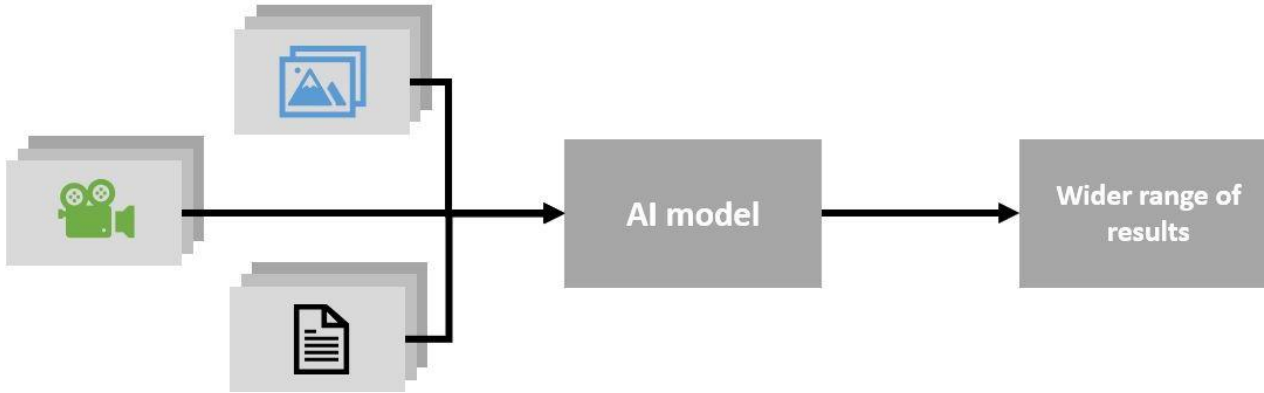
Better World Model and Knowledge
→ Robustness

I- Introduction

Unimodal AI model



Multimodal AI model

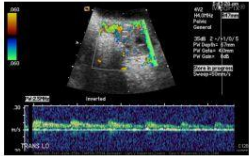
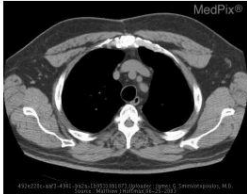
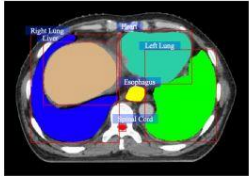
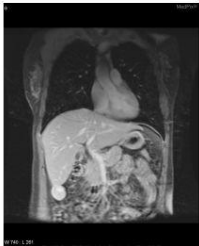


autonomous driving, image generation, robot navigation, ...

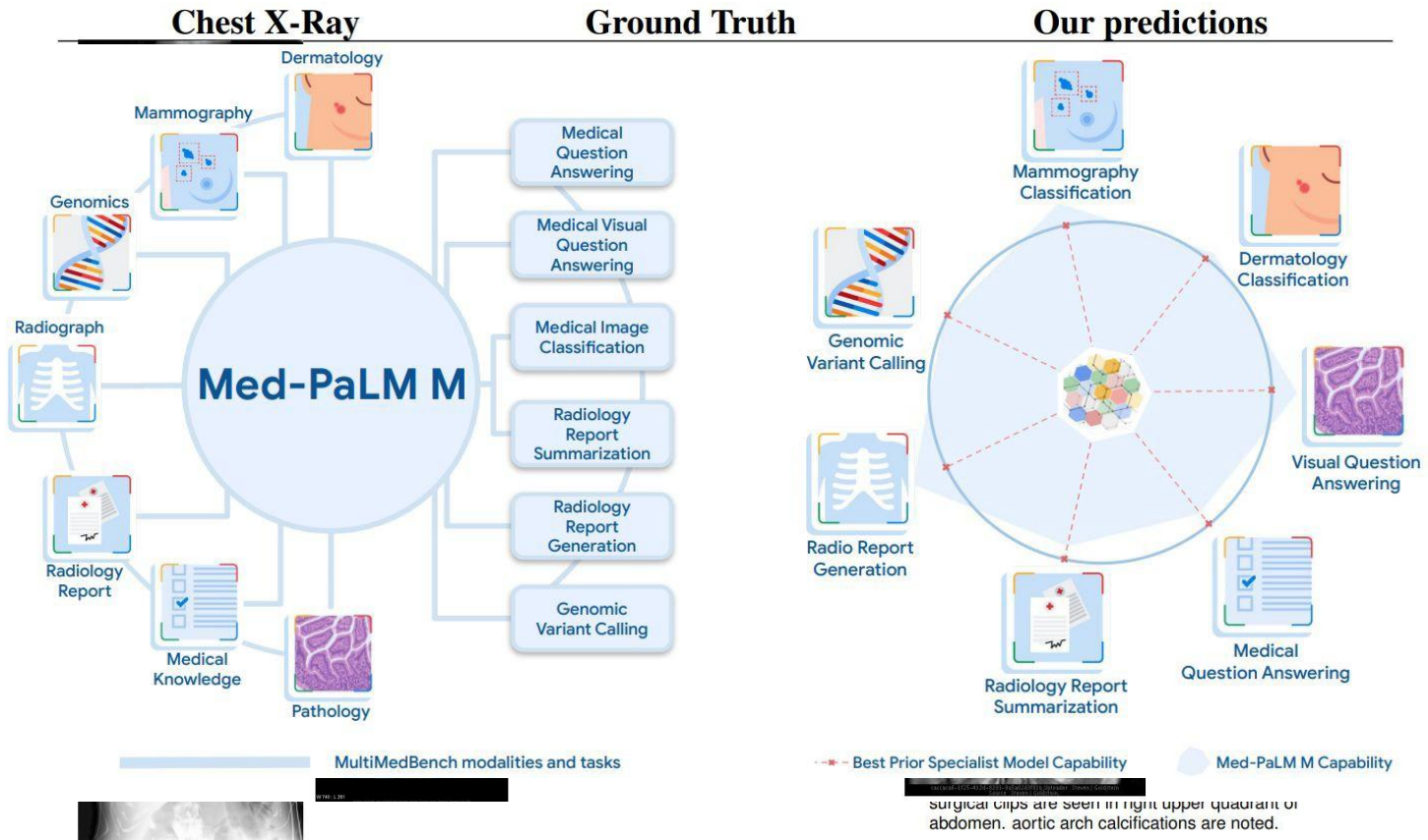
I- Introduction

Chest X-Ray	Ground Truth	Our predictions
	Lungs remain well inflated without evidence of focal airspace consolidation, pleural effusions, pulmonary edema or pneumothorax. Irregularity in the right humeral neck is related to a known healing fracture secondary to recent fall. PA and lateral views of the chest _____ at 09:55 are submitted.	no findings. no pneumonia. no pleural effusion. no edema. there is little change and no evidence of acute cardiopulmonary disease. no pneumonia, vascular congestion, pleural effusion. of incidental note is an azygos fissure, of no clinical significance. this raises possibility of a normal variant.
	1. Stable bilateral small pleural effusions and atelectasis. 2. Enlarged pulmonary artery, suggesting pulmonary hypertension. No significant interval change. Bilateral small pleural effusions and adjacent atelectasis are overall unchanged. The heart is top-normal in size, unchanged. The pulmonary artery is enlarged, suggesting pulmonary hypertension. No demand, focal consolidation to suggest pneumonia, or pneumothorax.	pleural effusion present. lung opacity present. no edema. cardiomegaly present. atelectasis present. as compared to previous radiograph, there is an increase in extent of a pre existing small left pleural effusion with subsequent atelectasis at left lung bases. otherwise, radiograph is unchanged. moderate cardiomegaly. mild fluid overload no overt pulmonary edema. no new focal parenchymal opacities suggesting pneumonia. unchanged position of right pectoral port a cath.
	There is decrease in now small right pleural effusion. There is no pneumothorax. There is a new right pacer pigtail catheter. Cardiomeastinal contours are unchanged. Lines and tubes are in standard position. Left lower lobe opacities, a combination of pleural effusion and atelectasis, are unchanged.	uncertain pneumonia. pleural effusion present. lung opacity present. atelectasis present. bilateral pleural effusions, left greater than right. bibasilar opacities potentially atelectasis in setting of low lung volumes. infection be excluded. frontal and lateral views of chest demonstrate low lung volumes, which accentuate bronchovascular markings. there are small bilateral pleural effusions, right greater than left, with adjacent atelectasis. there is no focal consolidation pneumothorax. cardiomeastinal silhouette is within normal limits. surgical clips are seen in right upper quadrant of abdomen. aortic arch calcifications are noted.

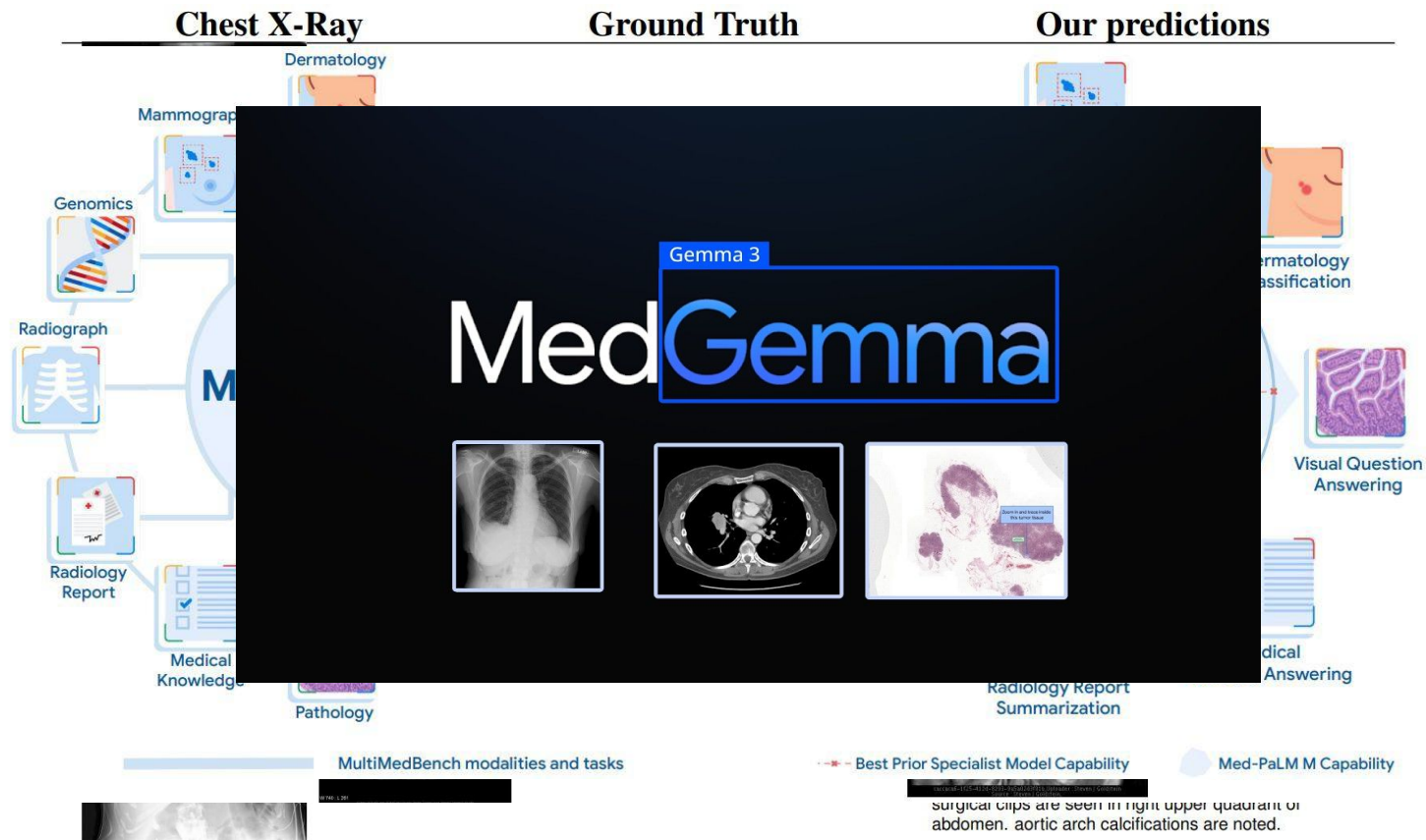
I- Introduction

	Chest X-Ray	Ground Truth	Our predictions
VQA-Med-2020 [13]		 Q: what abnormality is seen in the image? A: ovarian torsion	 Q: what is abnormal in the ct scan? A: partial anomalous pulmonary venous return
SLAKE [57]		Q: Does the image contain left lung? A: Yes	Q: What is the function of the rightmost organ in this picture? A: Breathe
VQA-Med-2021 [15]		Q: What is most alarming about this mri? A: focal nodular hyperplasia	 Q: What abnormality is seen in the image? A: Enhancing lesion right parietal lobe with surrounding edema
			surgical clips are seen in right upper quadrant of abdomen. aortic arch calcifications are noted.

I- Introduction



I- Introduction

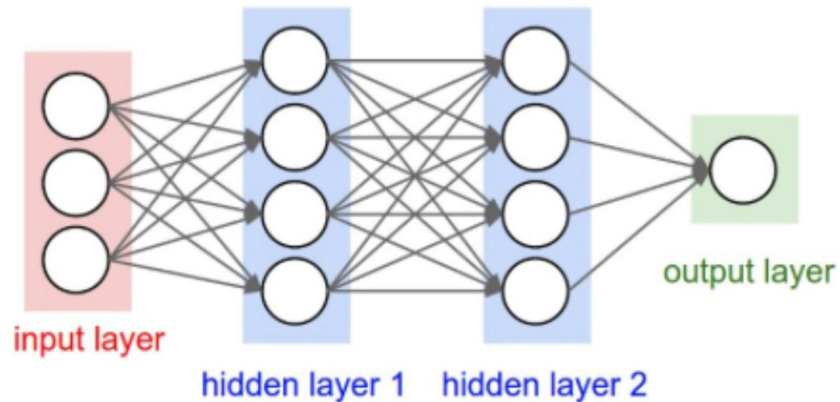


Recall

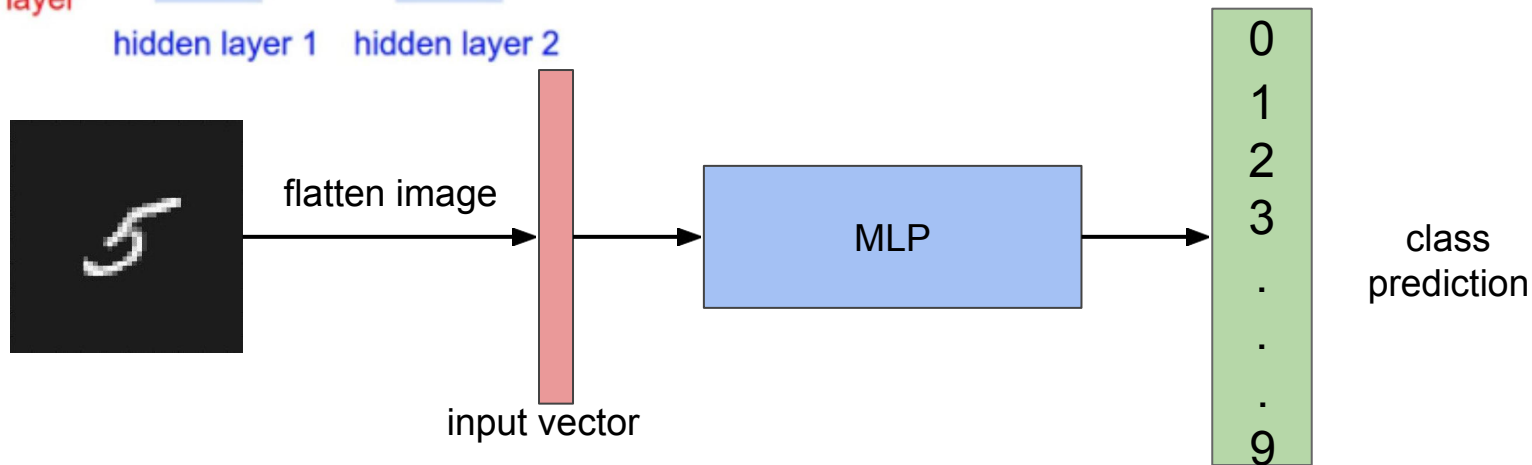
II- Recall

MLP :

- Most basic Deep Learning Model that can handle only **vector** as input



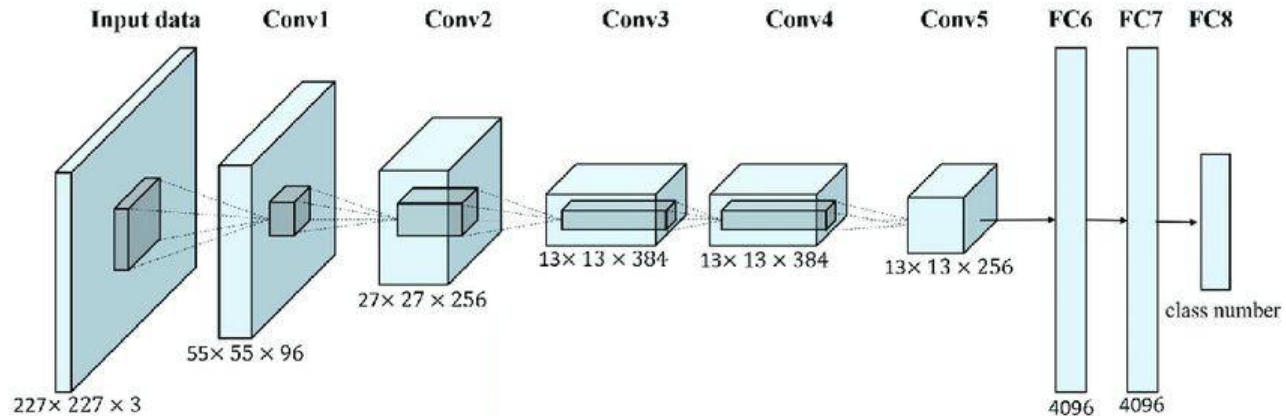
Linear Layer : $\hat{y} = f(Wx + b)$



II- Recall

CNN :

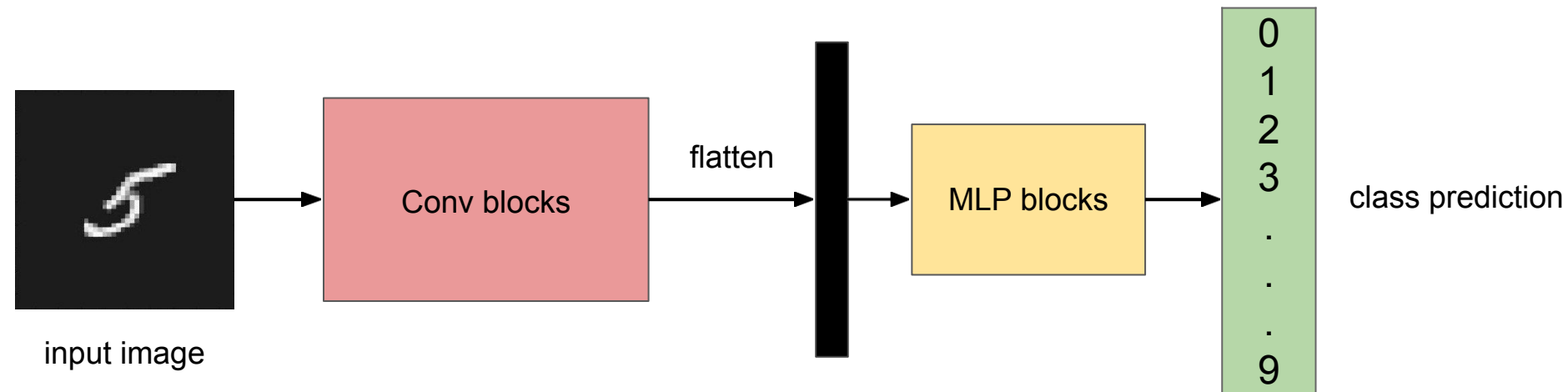
- Very useful to treat **images** or input which have spatial relation thanks to Convolutional Layers



II- Recall

CNN :

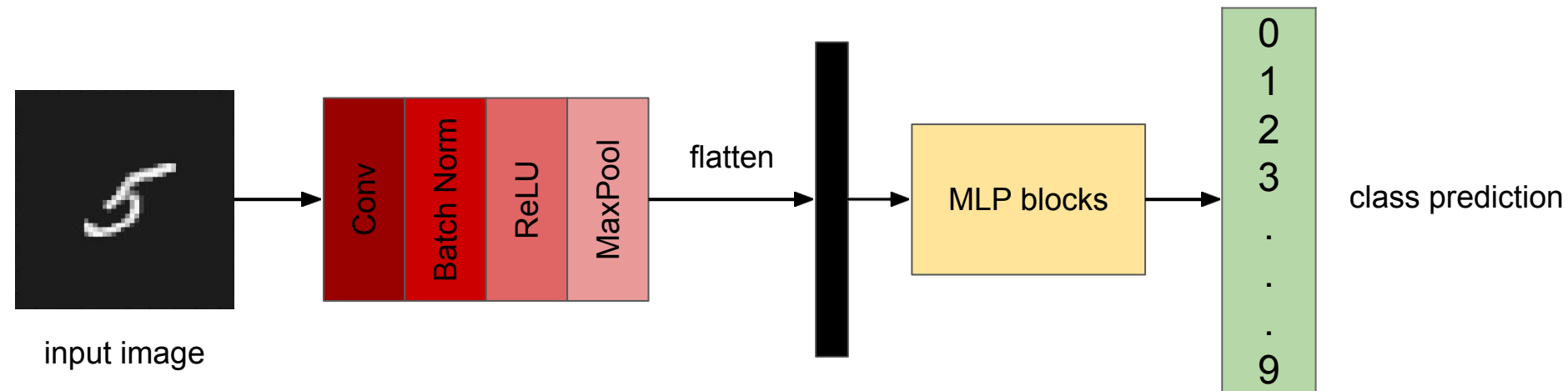
- Very useful to treat **images** or input which have spatial relation thanks to Convolutional Layers



II- Recall

CNN :

- Very useful to treat **images** or input which have spatial relation thanks to Convolutional Layers



II- Recall

RNN :

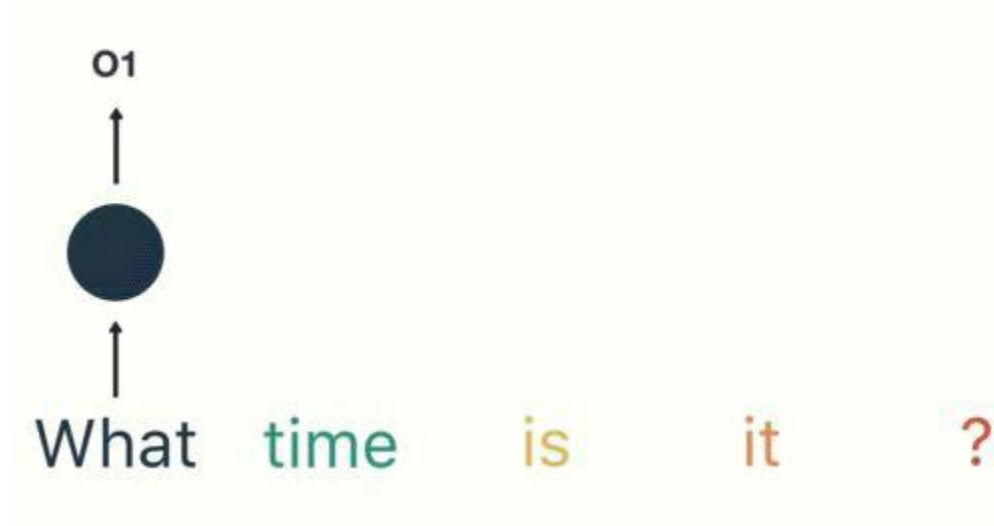
- Very good to treat **sequential data** thanks to its hidden state acting like a memory

What time is it ?

II- Recall

RNN :

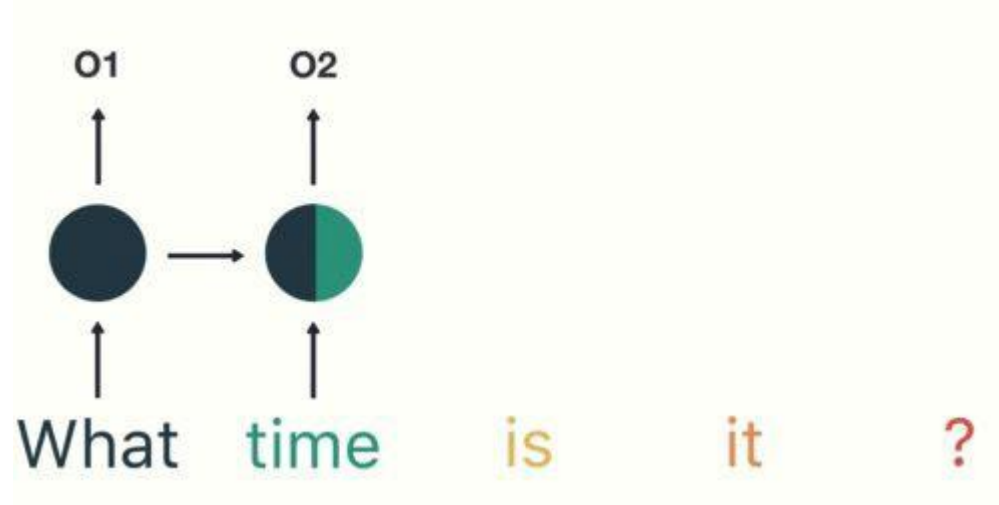
- Very good to treat **sequential data** thanks to its hidden state acting like a memory



II- Recall

RNN :

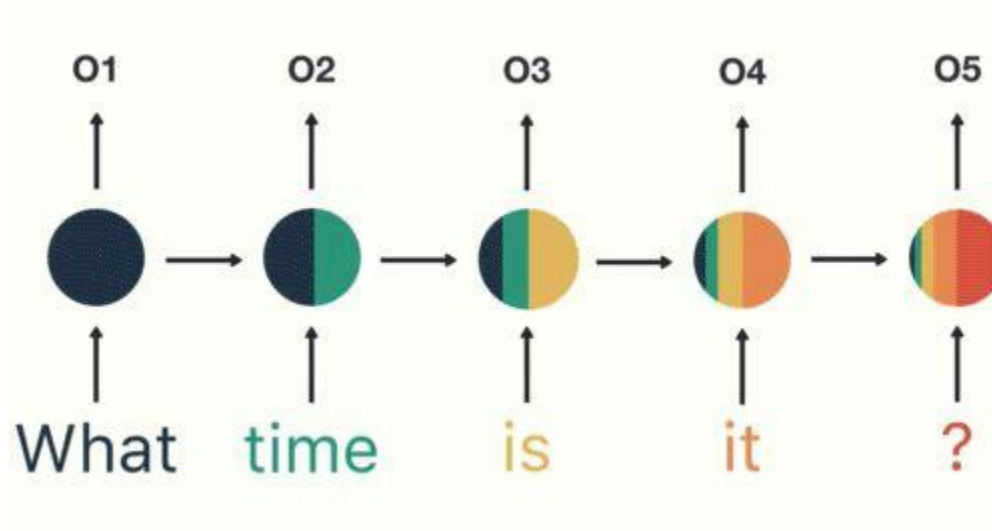
- Very good to treat **sequential data** thanks to its hidden state acting like a memory



II- Recall

RNN :

- Very good to treat **sequential data** thanks to its hidden state acting like a memory

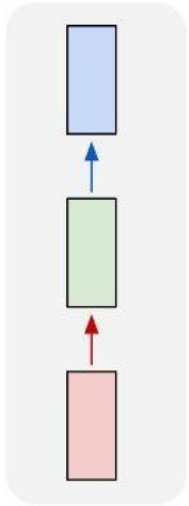


II- Recall

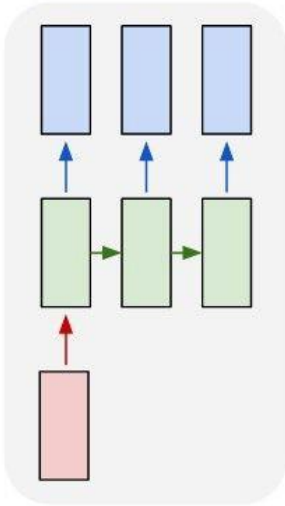
RNN :

- Different type of RNN depending on the task you want

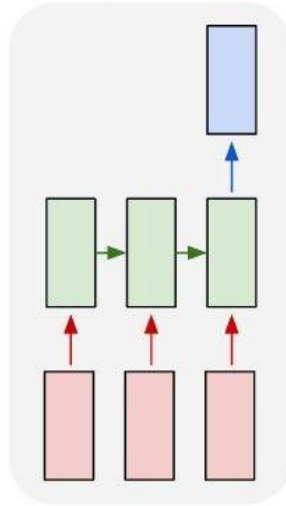
one to one



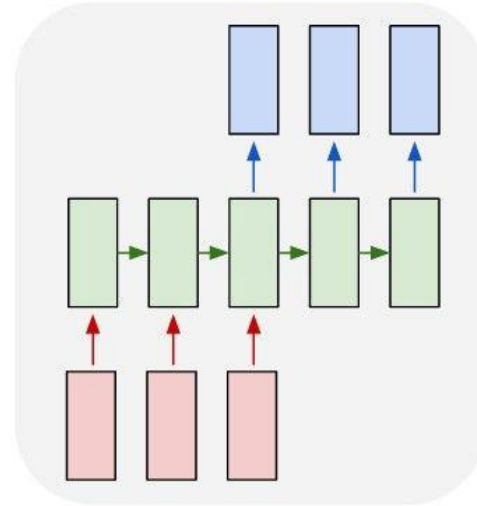
one to many



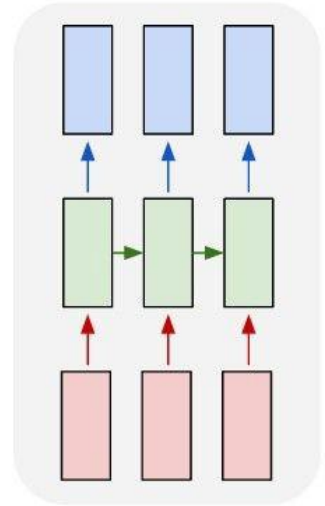
many to one



many to many



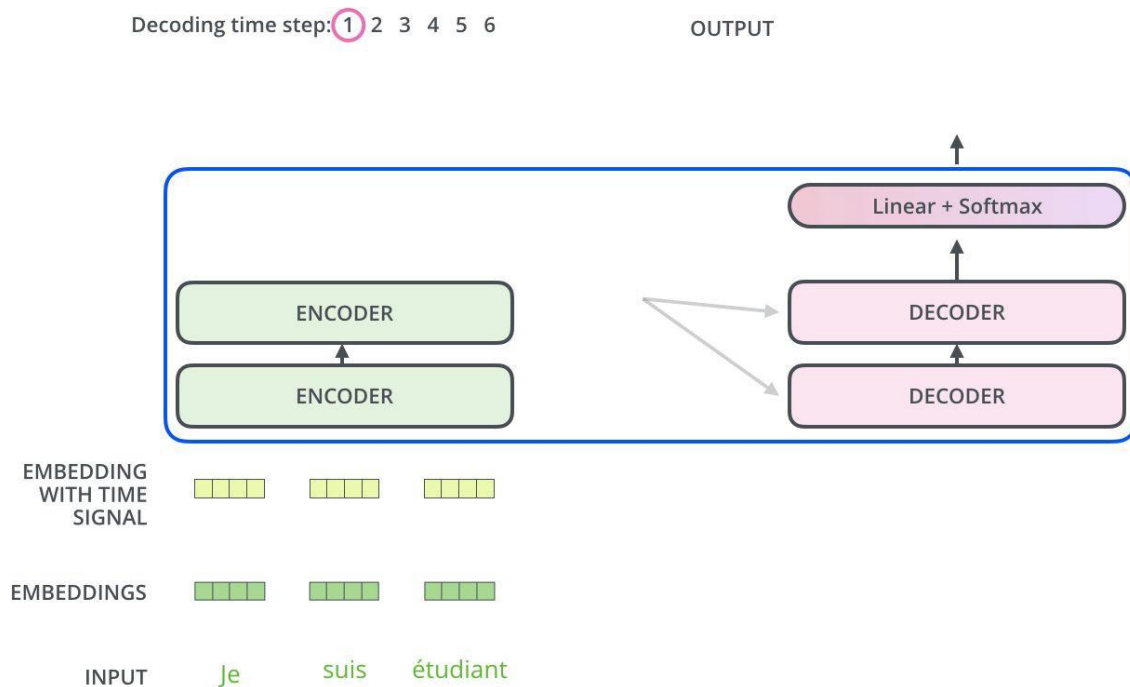
many to many



II- Recall

Transformers :

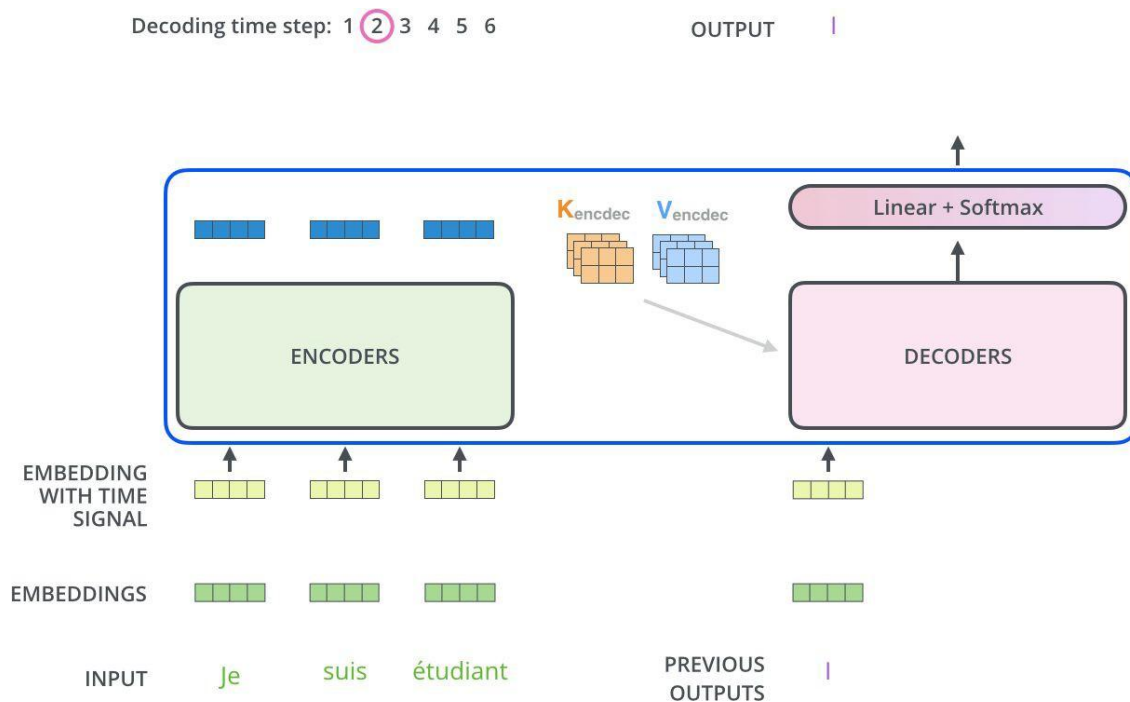
- Very good to treat **sequential data** (ex : text)
- Able to learn interactions between different element of the sentence



II- Recall

Transformers :

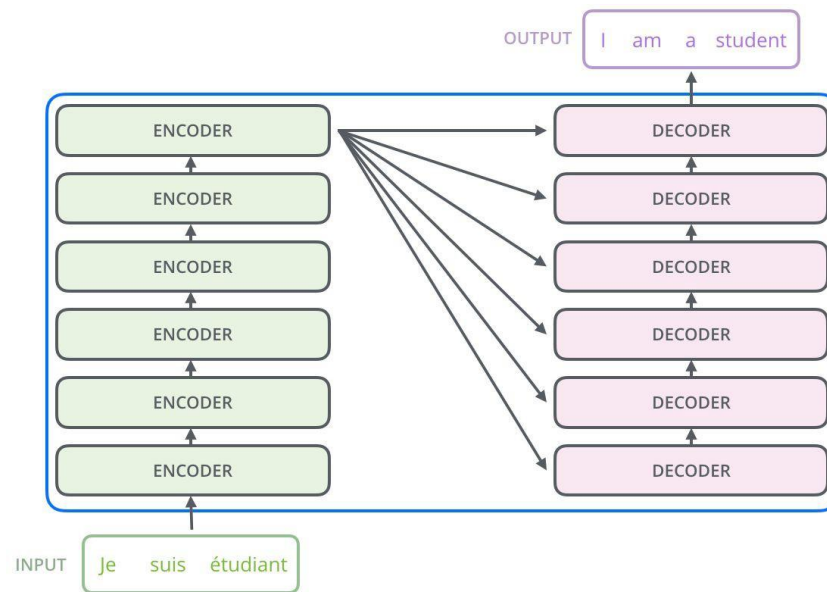
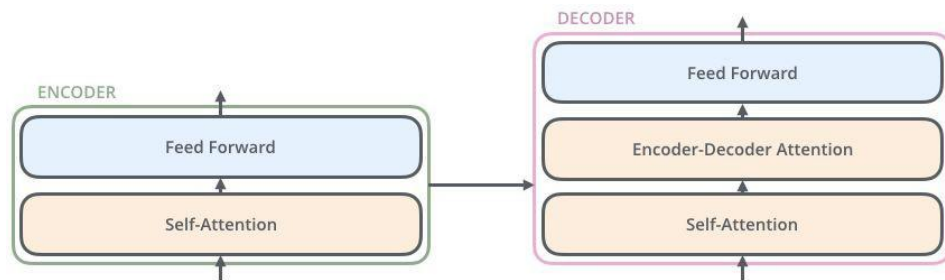
- Very good to treat **sequential data** (ex : text)
- Able to learn interactions between different element of the sentence



II- Recall

Transformers :

what's inside encoder and decoder ?

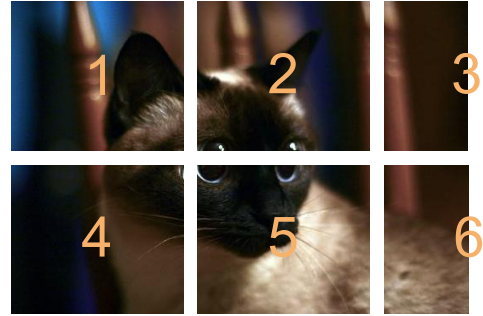


Global architecture Transformer

II- Recall

Transformers :

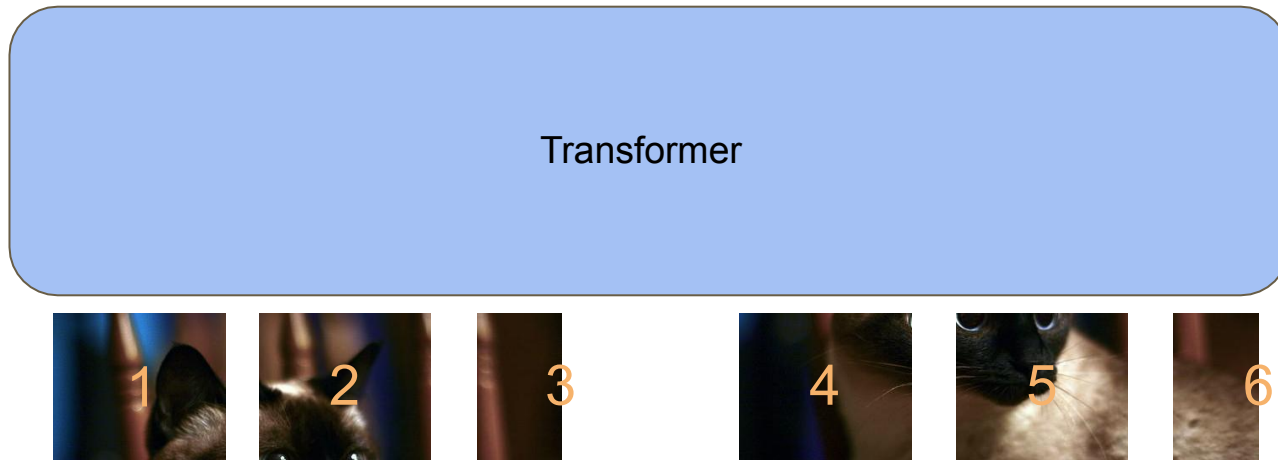
- Now also used for images → need to convert image to sequence



II- Recall

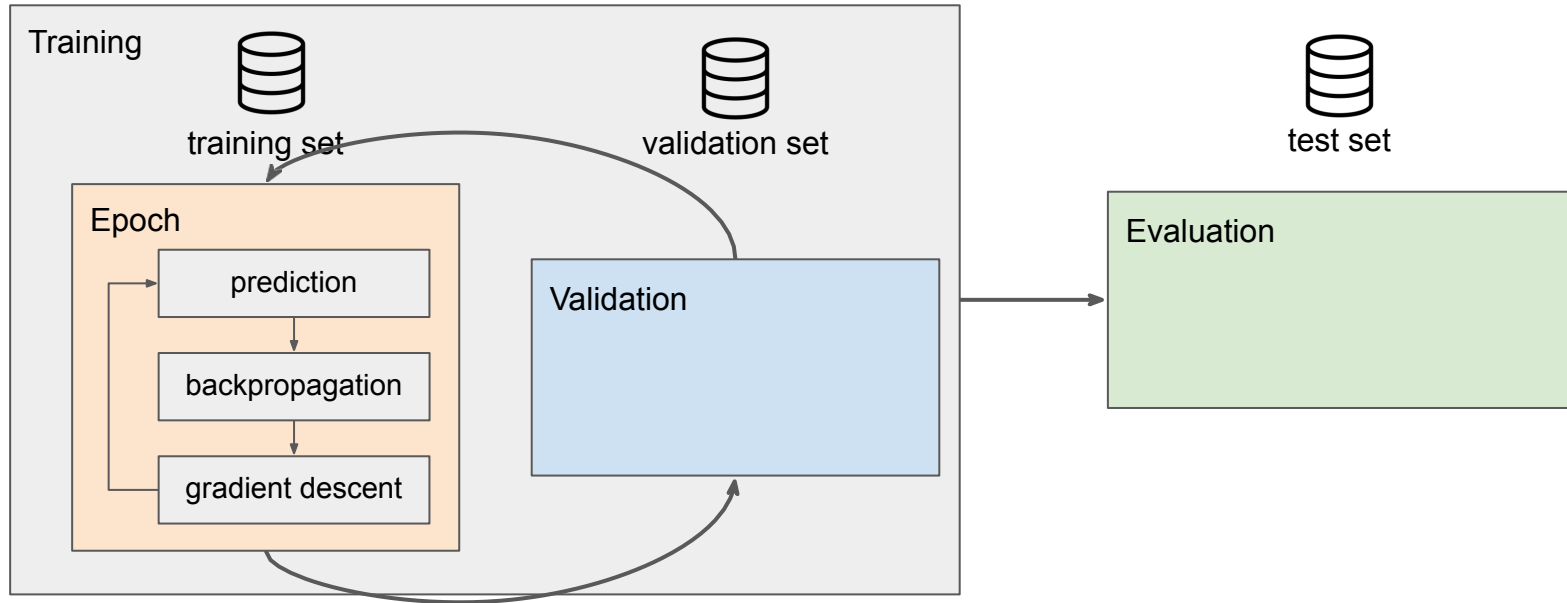
Transformers :

- Now also used for images → need to convert image to sequence
- Image divide into patches passed to the Transformer



II- Recall

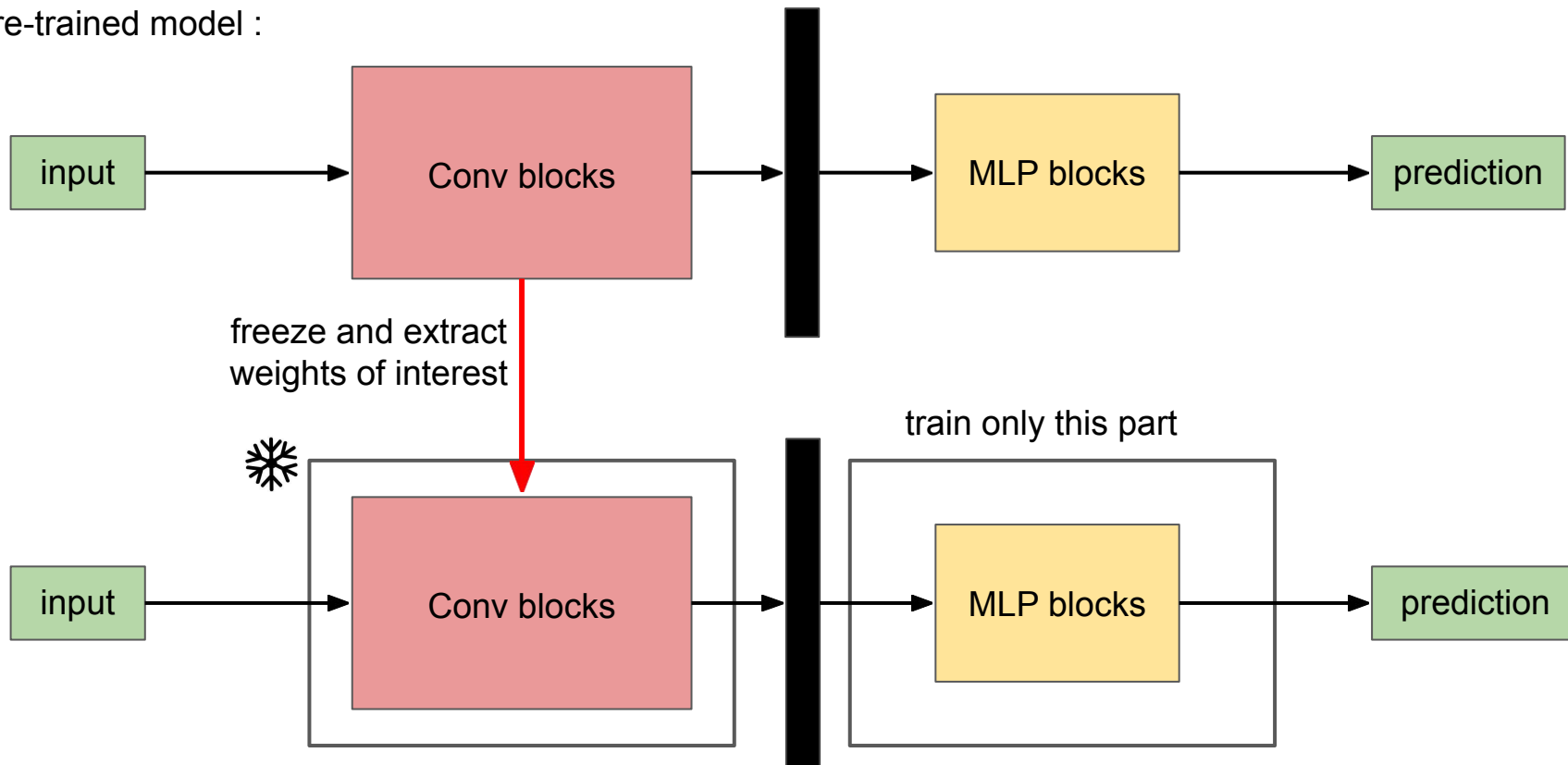
Train a model :



II- Recall

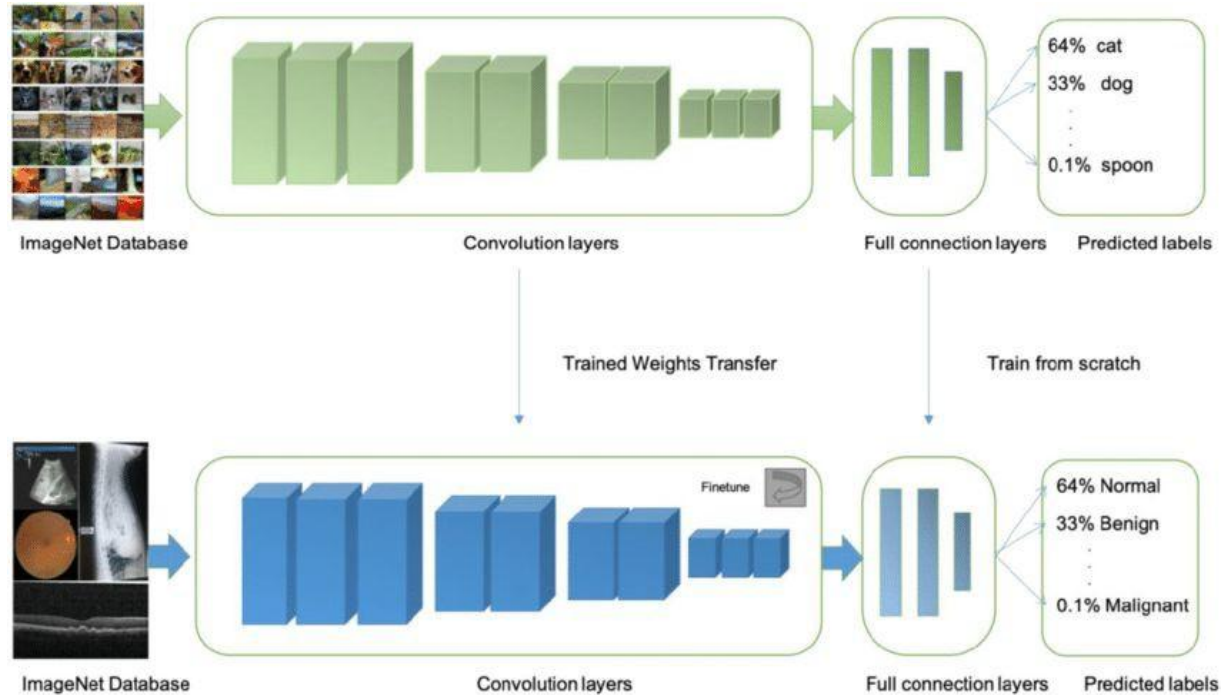
Use pre-trained model : **transfer learning**

Pre-trained model :



II- Recall

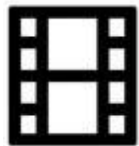
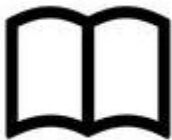
Use pre-trained model :



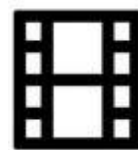
How to learn from Multimodality ?

How to learn from Multimodality ?

Unimodal Data



Multimodal Data



How to learn from Multimodality ?



Raw modalities :
close to the sensor



Abstract modalities :
far from the sensor

Speech signal

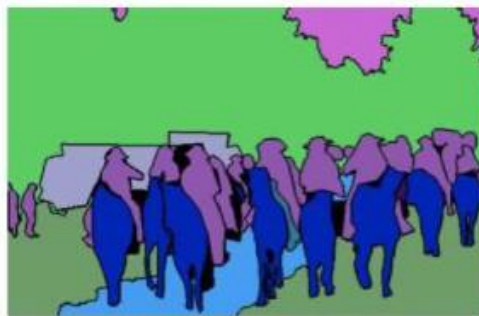
Language

Object categories

Image

Detected objects

Sentiment intensity



How to learn from Multimodality ?



the underside of a large
airplane in the sky.

a plane is taking off from the
airport.

an airplane flying in the air with
a sky background

an airplane in the air seen from
below at twilight

a grey jet airliner passing
across a hazy blue sky.

image + text : multimodal

How to learn from Multimodality ?



image + segmented image : multimodal

How to learn from Multimodality ?



the underside of a large
airplane in the sky.

a plane is taking off from the
airport.

an airplane flying in the air with
a sky background

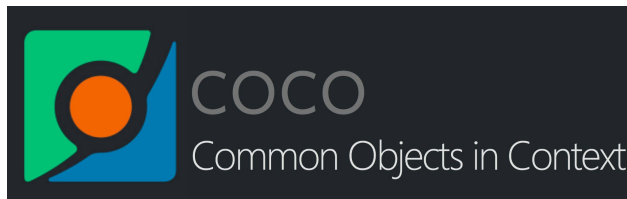
an airplane in the air seen from
below at twilight

a grey jet airliner passing
across a hazy blue sky.

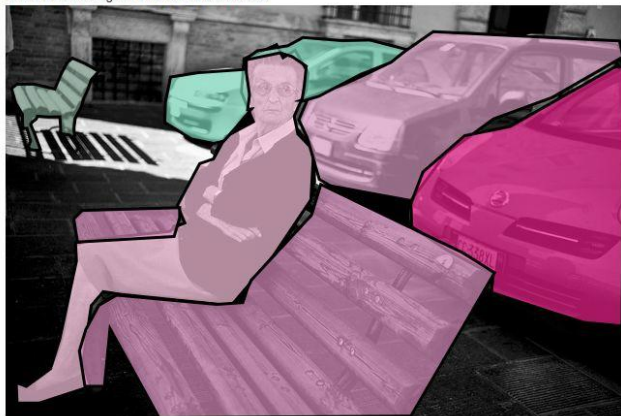
image + segmented image + text : multimodal

How to learn from Multimodality ?

Datasets



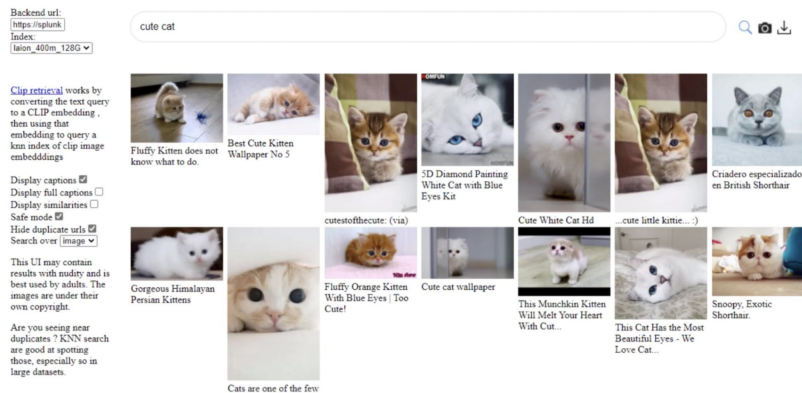
a woman sitting on a bench with cars behind her.
there is a woman sitting on a bench in front of cars
a woman that is sitting on a wooden bench.
black and white photo of a woman on a bench.
a woman is sitting on a wood bench outside



- 330K images
- Images segmentation
- Simple descriptive sentences

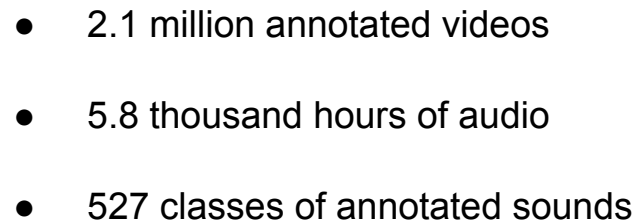
How to learn from Multimodality ?

Datasets



- Image text pairs
- 400M or 5B pairs
- Lot of different specialized version

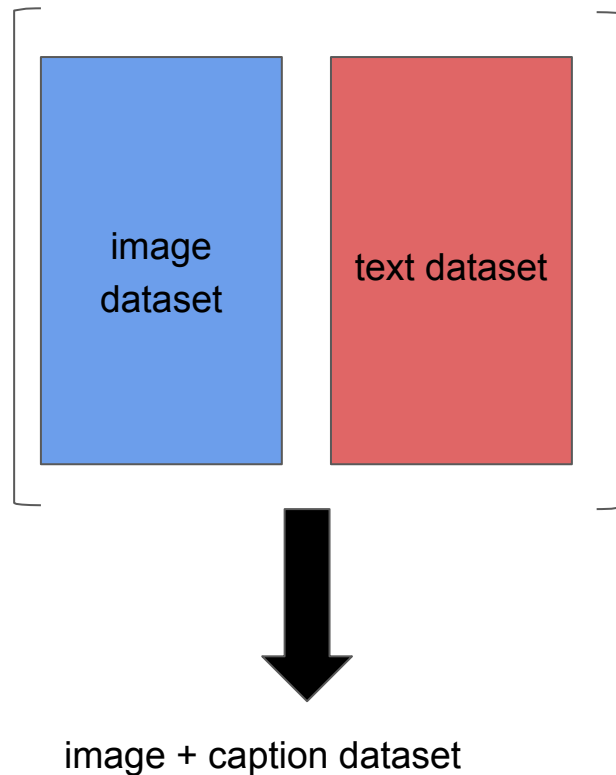
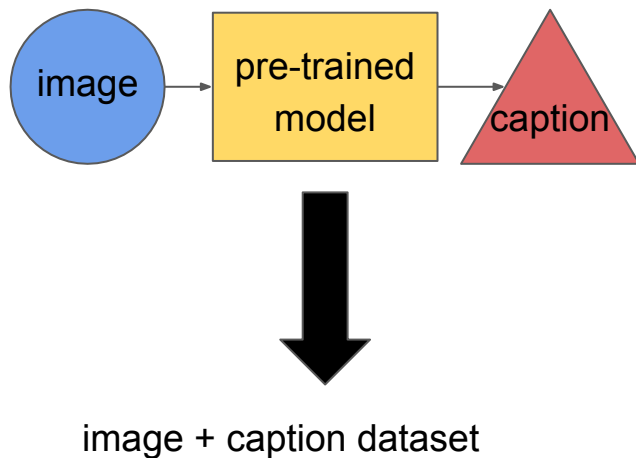
Datasets



How to learn from Multimodality ?

Datasets :

- create own multimodal dataset



How to learn from Multimodality ?

How to train a model that take as input different modalities ?

How to learn from Multimodality ?

Concatenation of raw modalities :

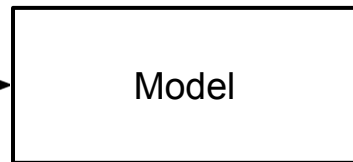
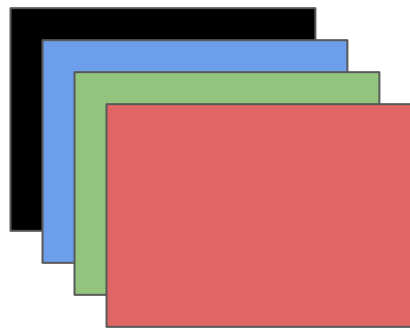
- Need to be very close (ex : 2 images : RGB + Depth)
- Increase the information given to the model to do better prediction



RGB



Depth

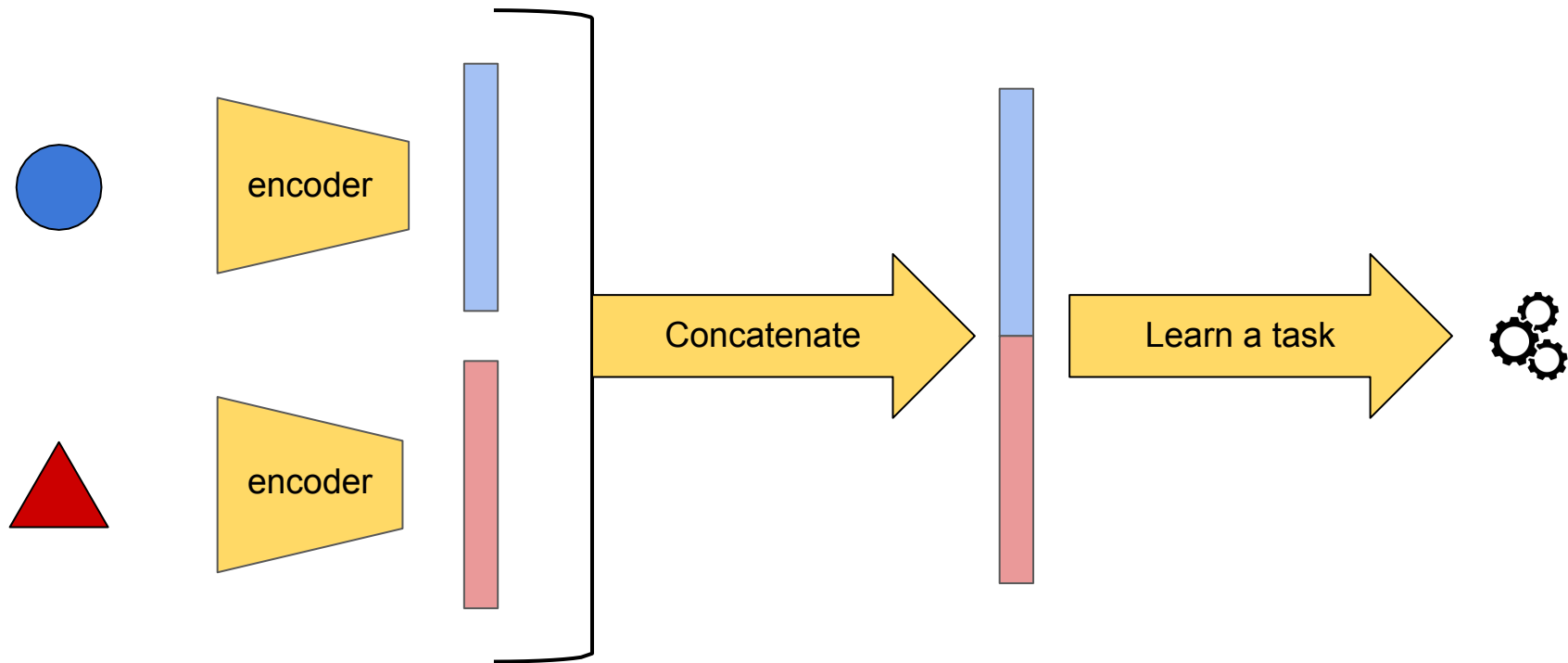


prediction

concatenation → 4 channels

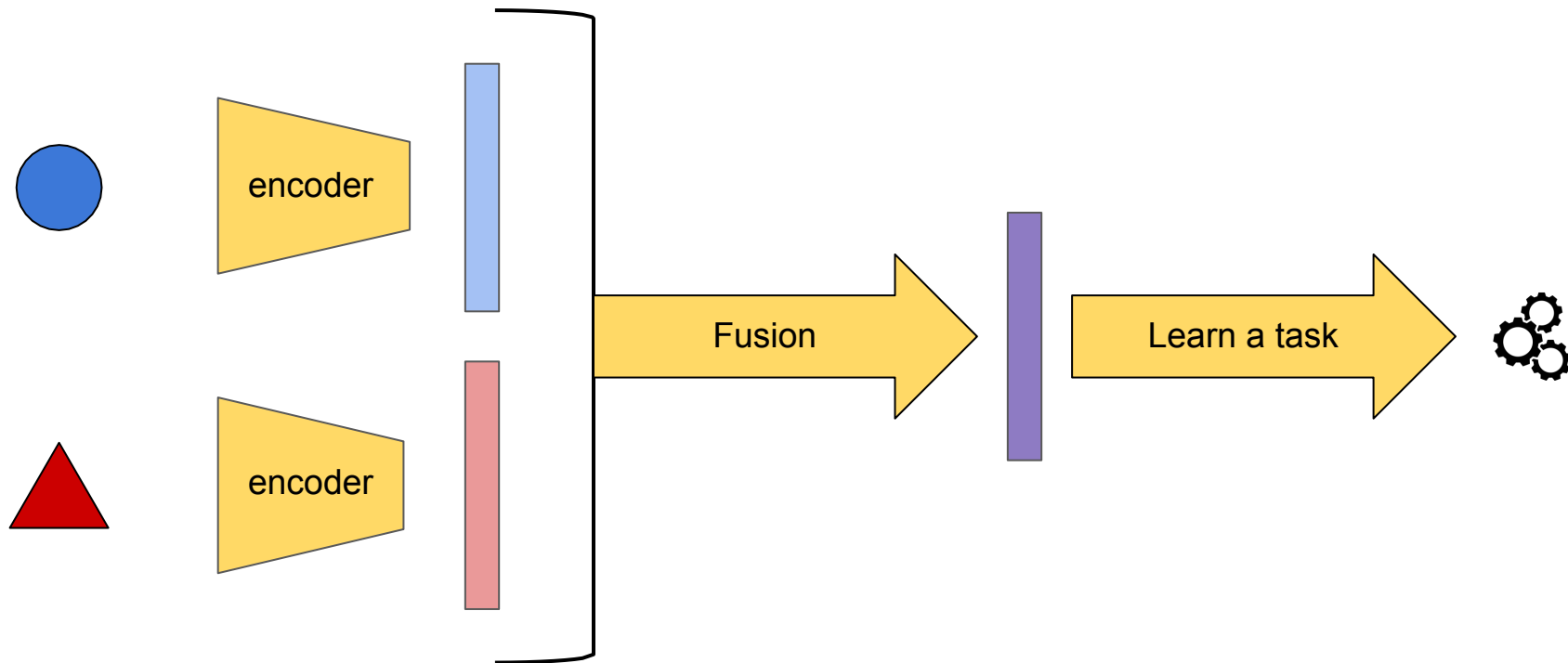
How to learn from Multimodality ?

Common approach : Encode modalities in latent space and merge them



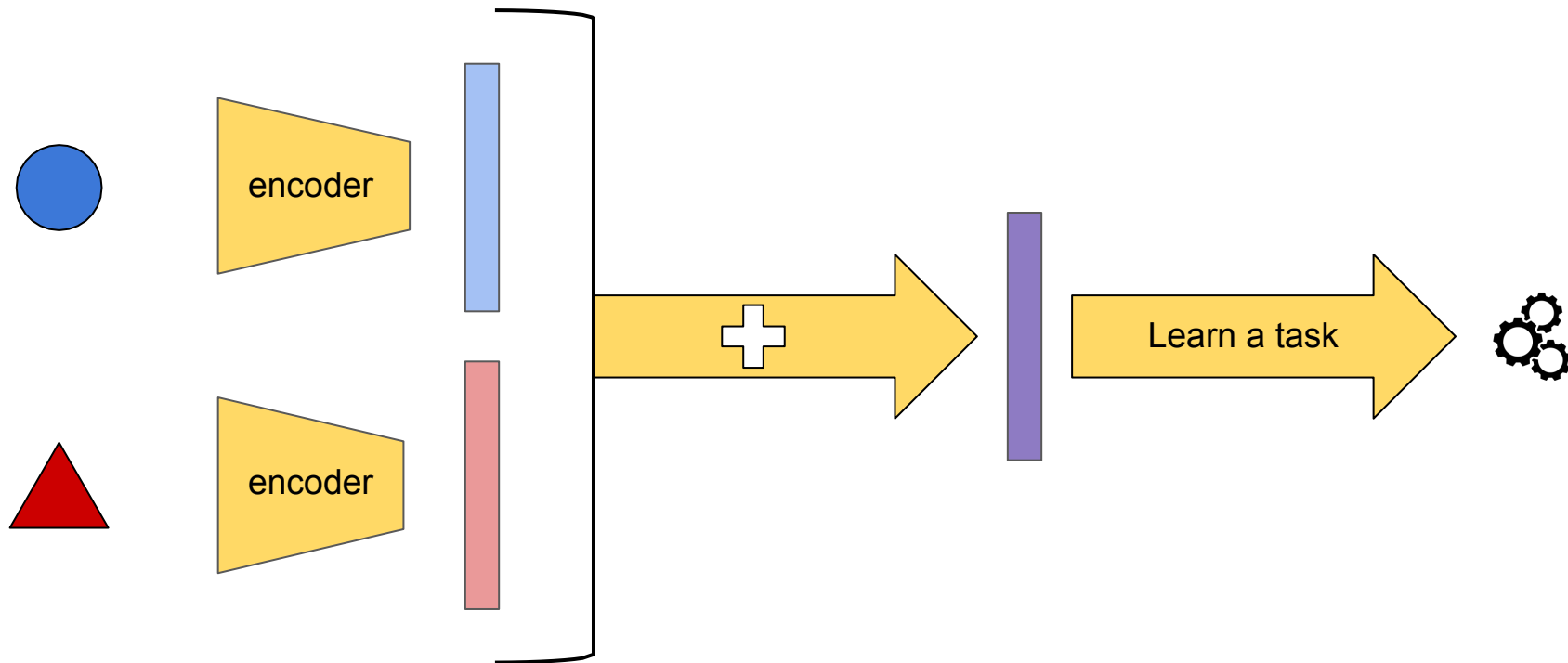
How to learn from Multimodality ?

Fusion of the latent vectors



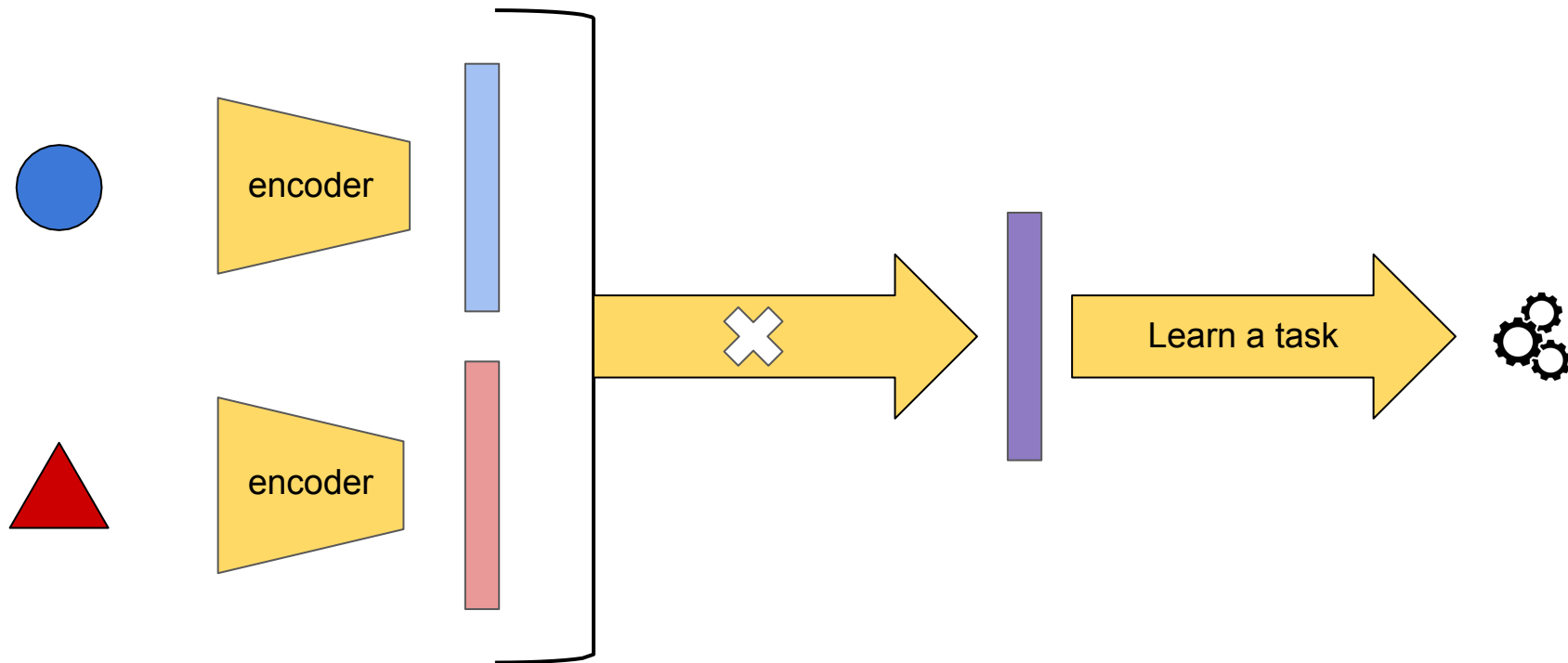
How to learn from Multimodality ?

Fusion of the latent vectors



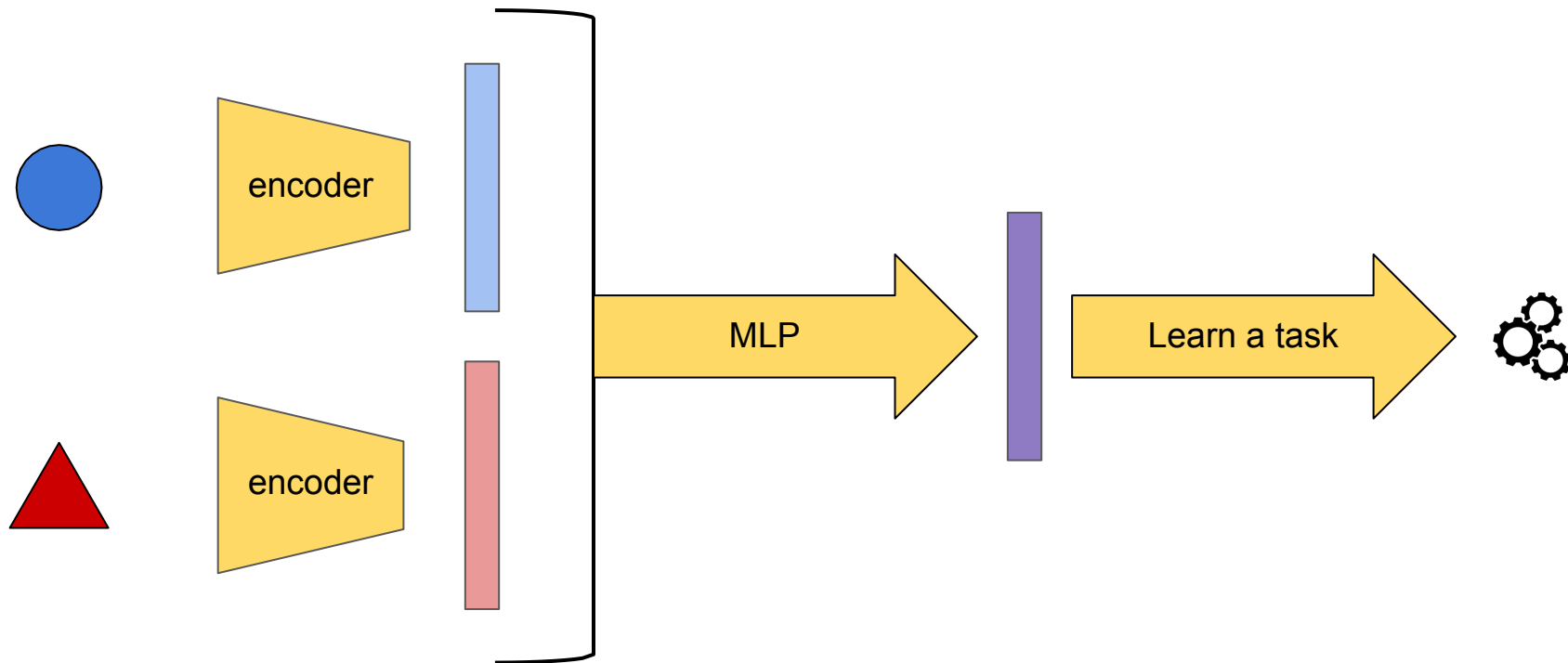
How to learn from Multimodality ?

Fusion of the latent vectors



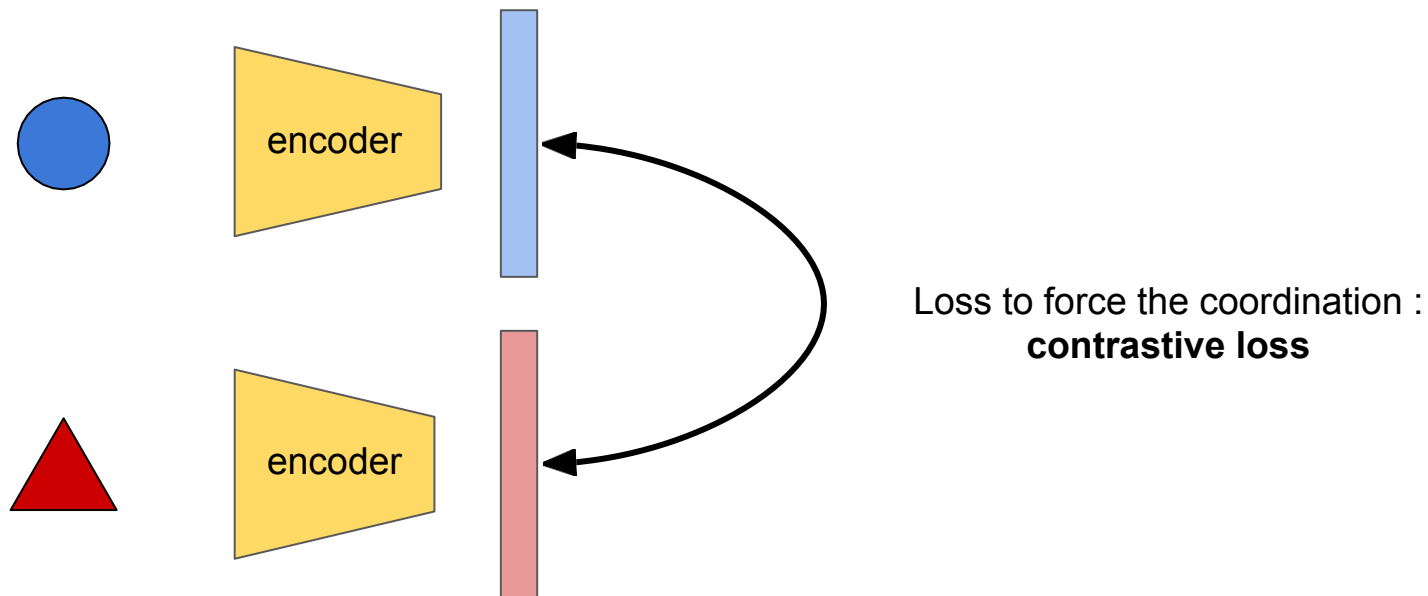
How to learn from Multimodality ?

Fusion of the latent vectors



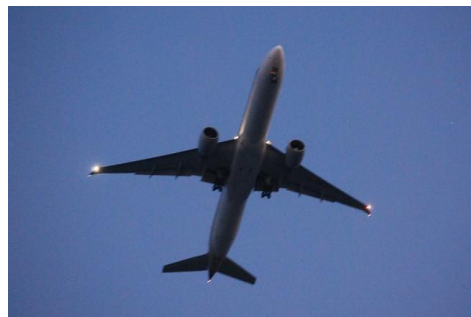
How to learn from Multimodality ?

Coordinate the latent vectors

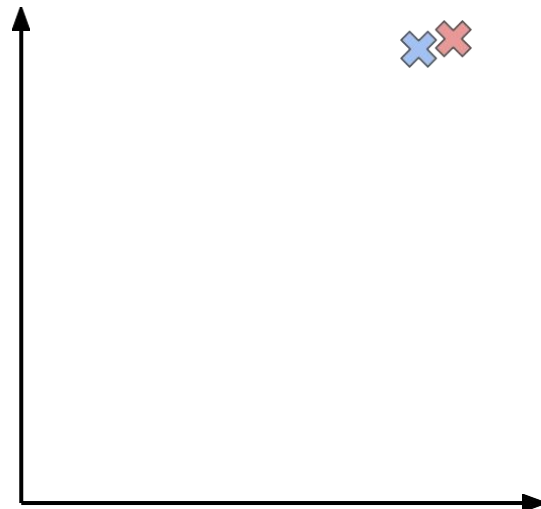
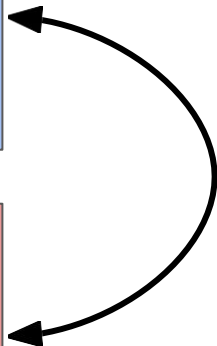
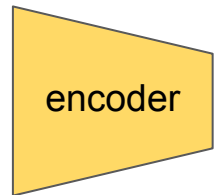
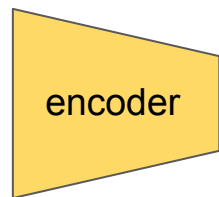


How to learn from Multimodality ?

Coordinate the latent vectors



a plane

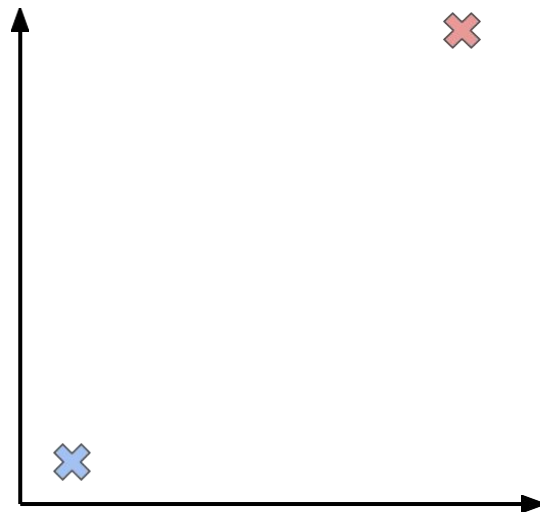
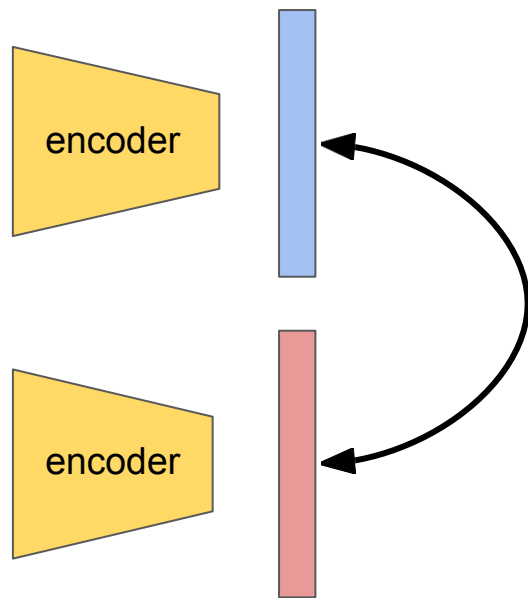


How to learn from Multimodality ?

Coordinate the latent vectors



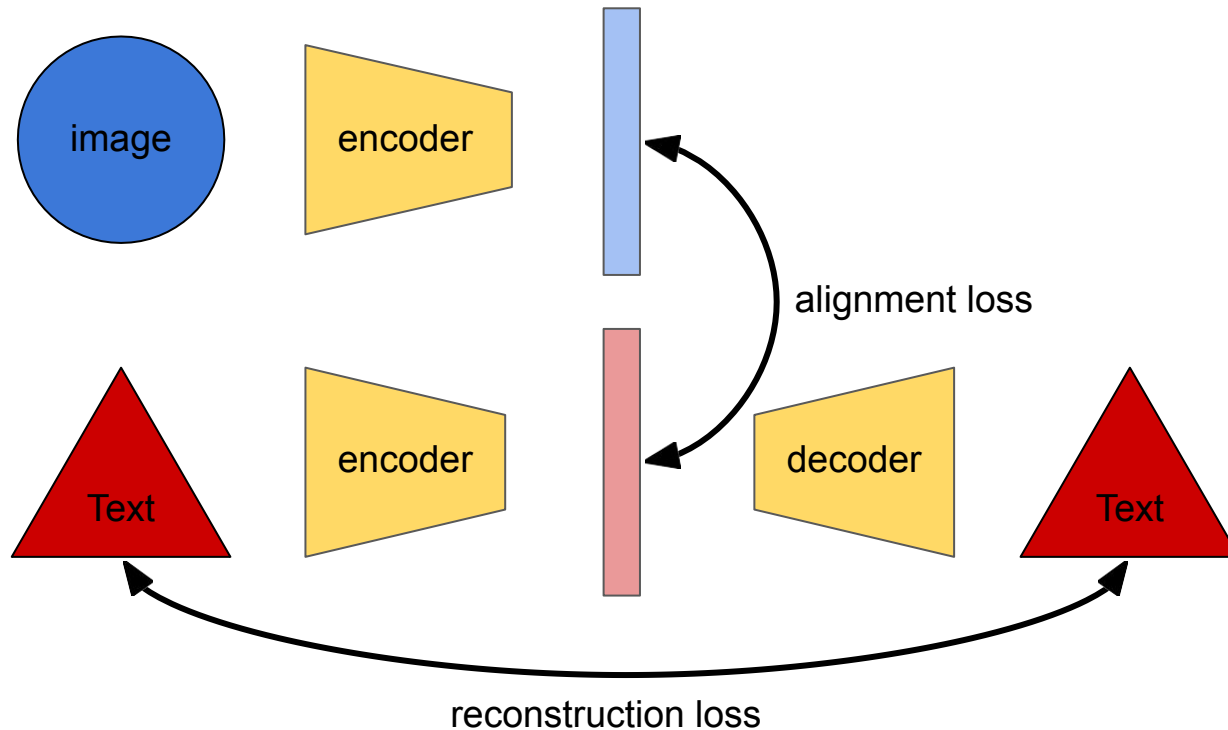
a plane



How to learn from Multimodality ?

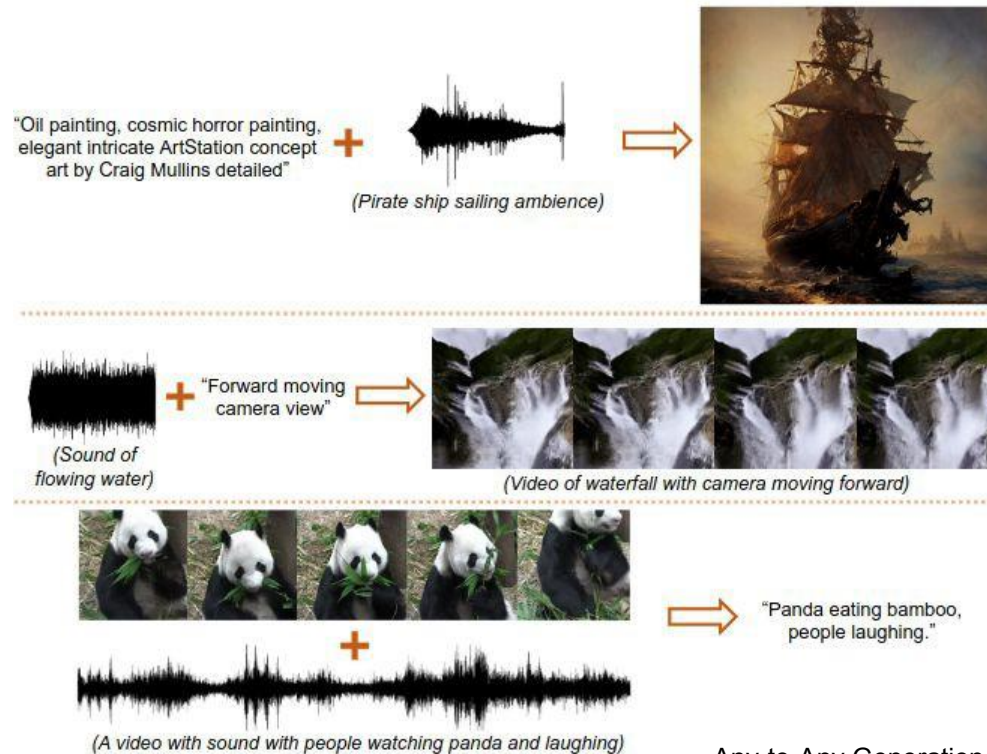
Coordinate the latent vectors

- example : learn a captioning model (create description of images)



How to learn from Multimodality ?

Coordinate the latent vectors

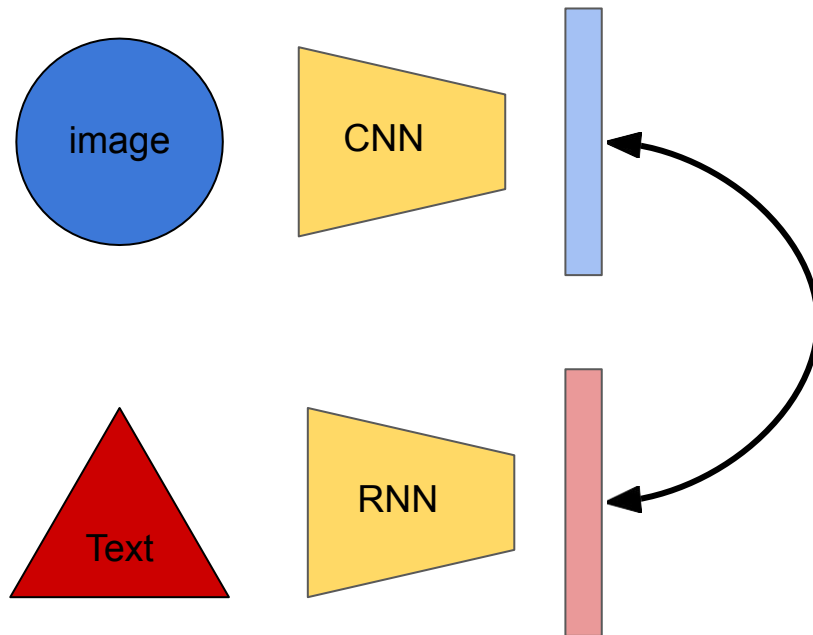


Any-to-Any Generation via Composable Diffusion; Tang & al.

How to learn from Multimodality ?

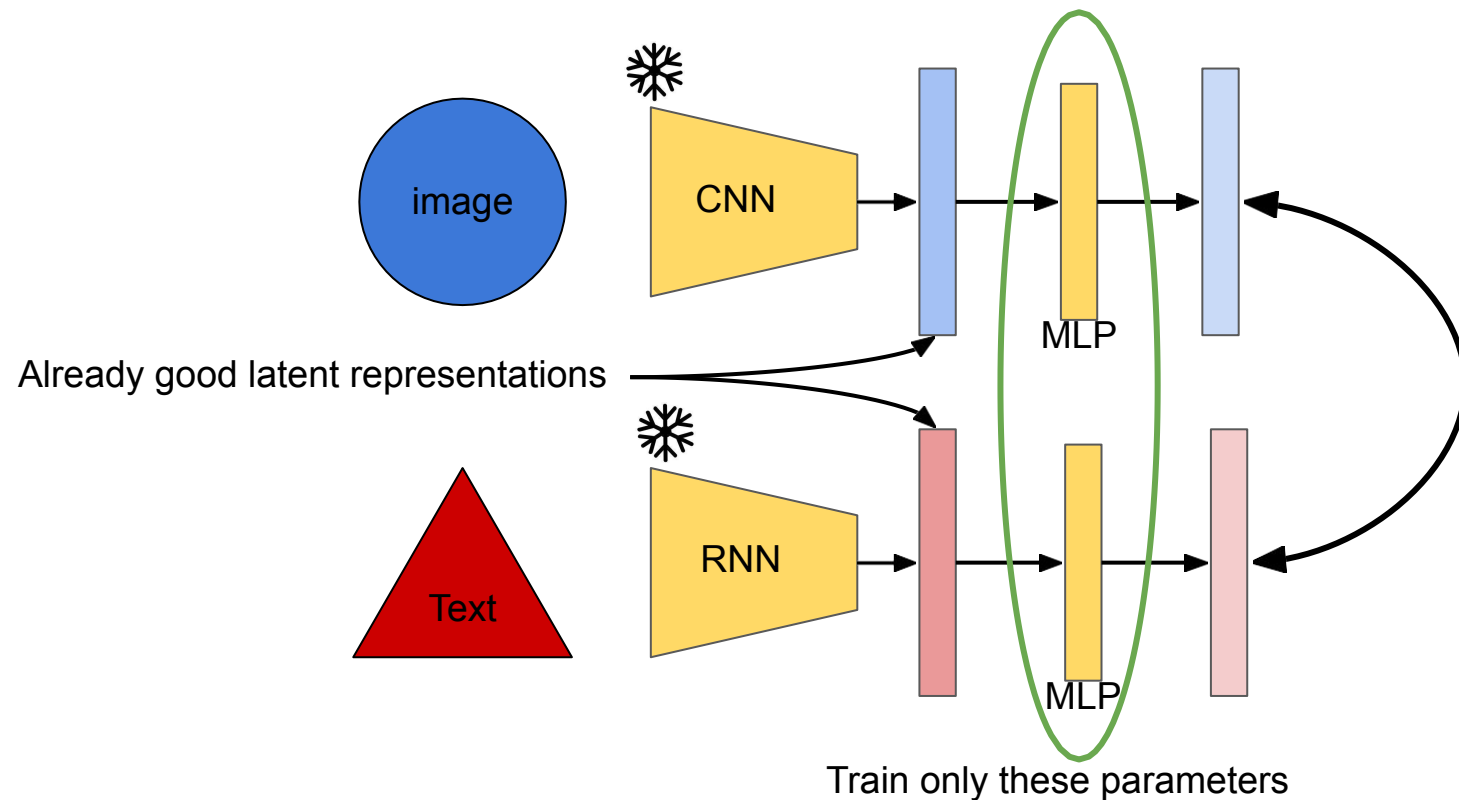
Learn directly the fusion or the coordination with the encoders:

- complex with a lot of parameters to train



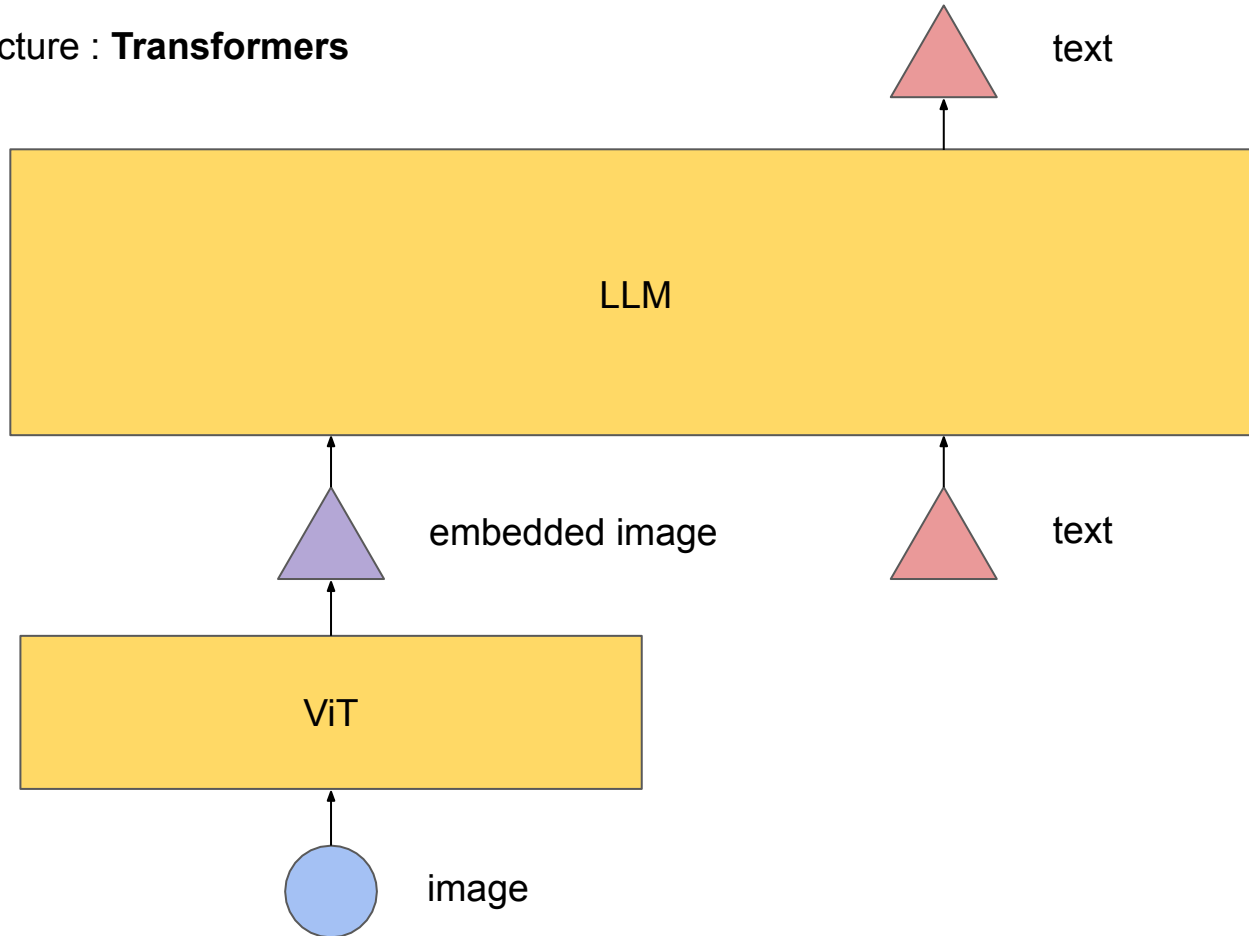
How to learn from Multimodality ?

Use pre-trained models to encode different modalities :



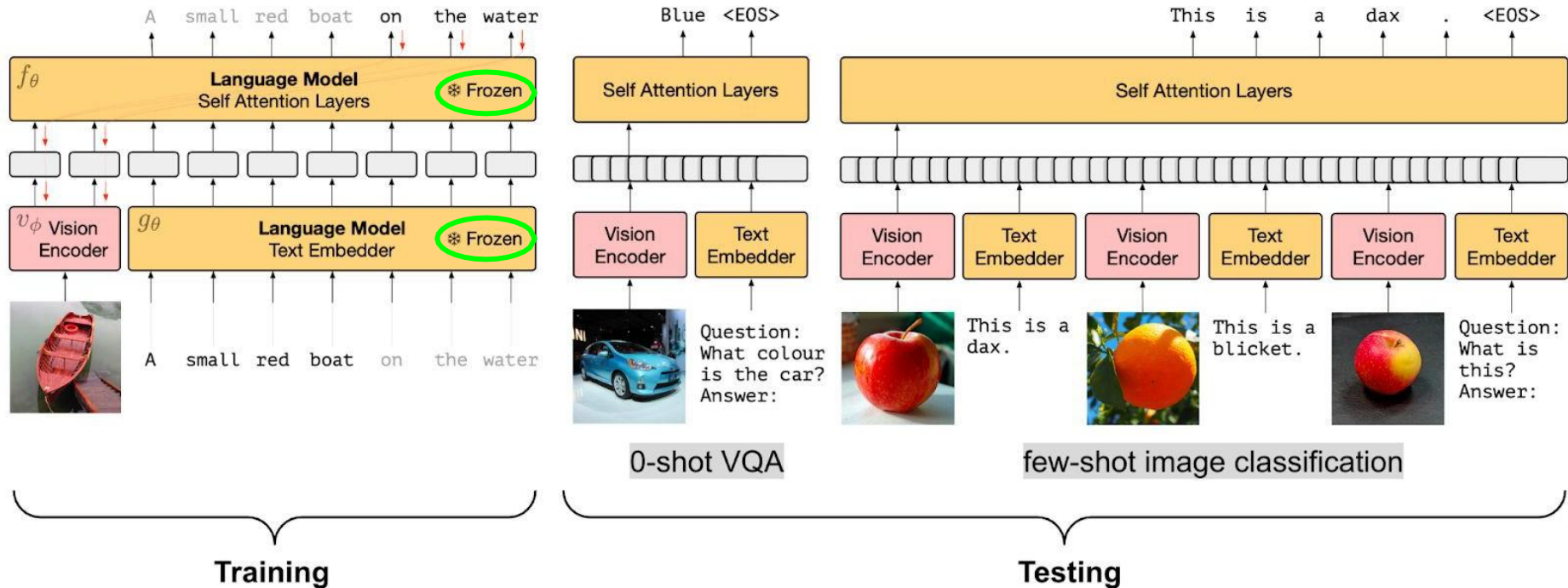
How to learn from Multimodality ?

Popular architecture : **Transformers**



How to learn from Multimodality ?

Popular architecture : **Transformers**



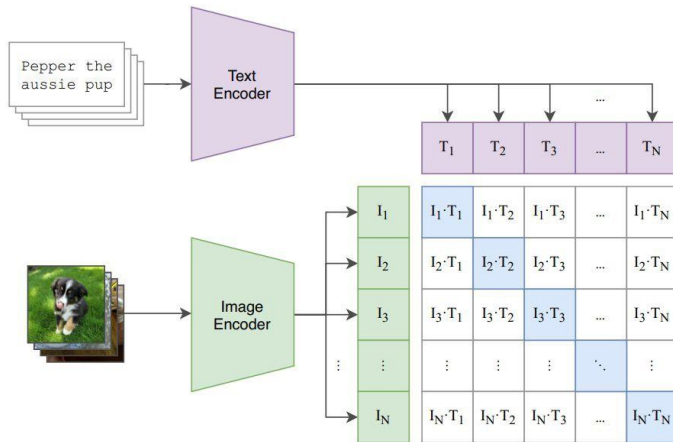
Multimodal Few Shot Learning with Frozen Language Model, Tsimpoukelli & al.

How to learn from Multimodality ?

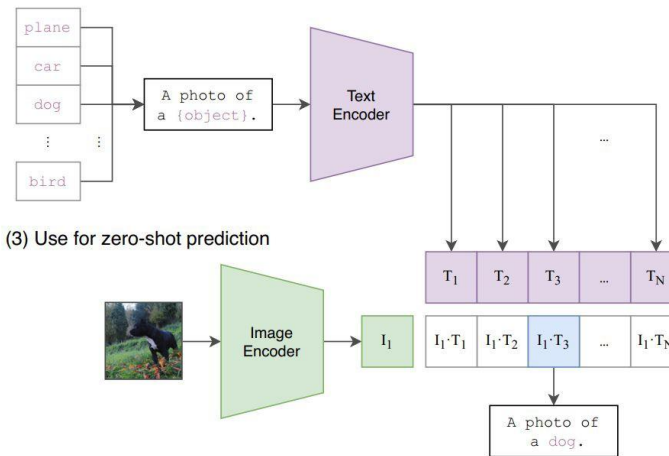
Popular architecture : **Foundation model**

CLIP : trained to encode image and text in aligned latent space

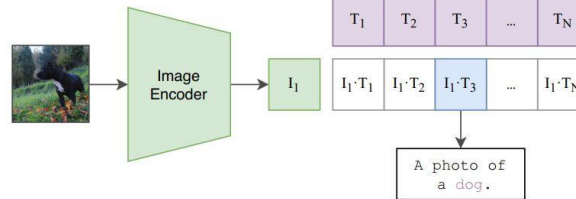
(1) Contrastive pre-training



(2) Create dataset classifier from label text



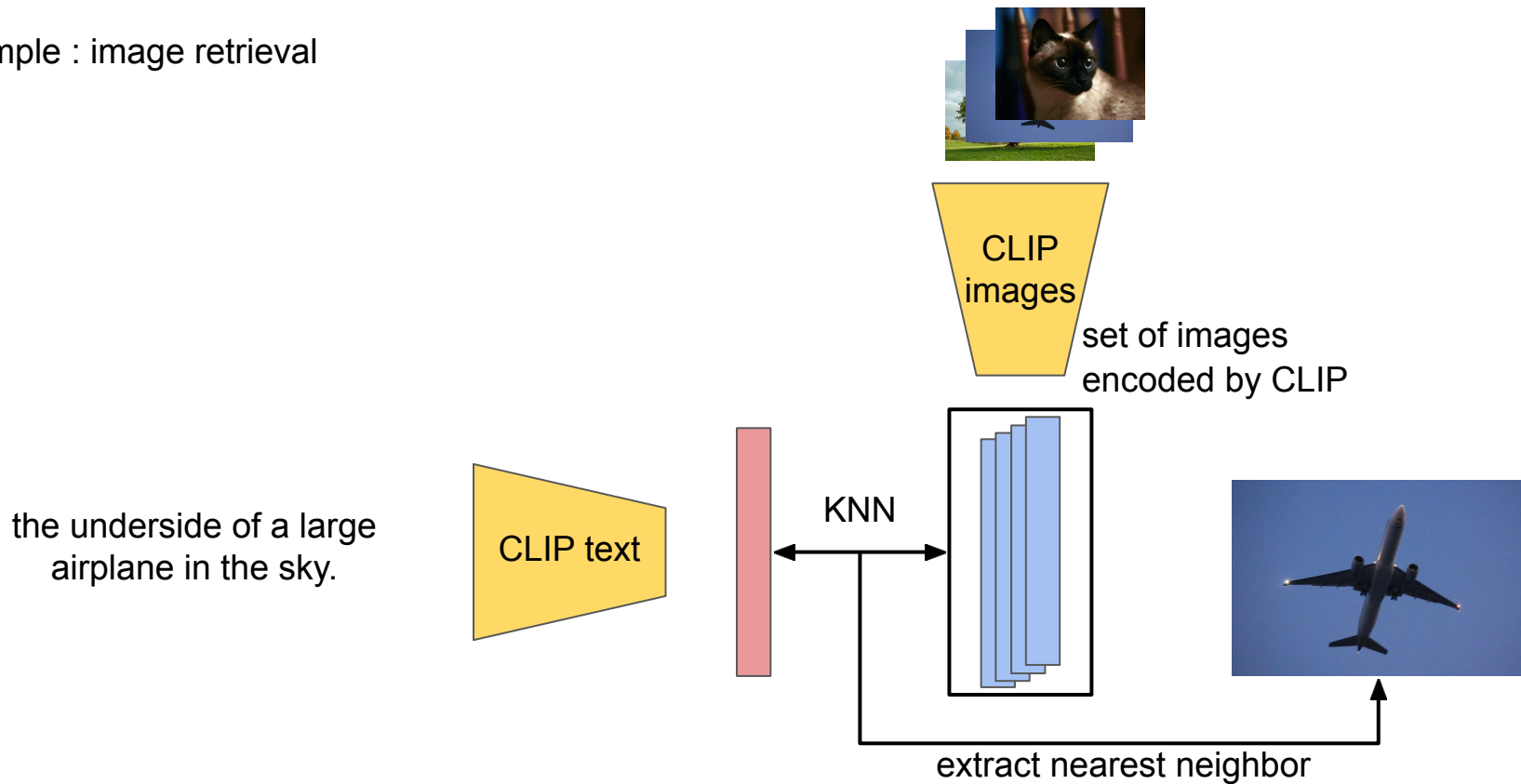
(3) Use for zero-shot prediction



How to learn from Multimodality ?

Popular architecture : **Foundation model**

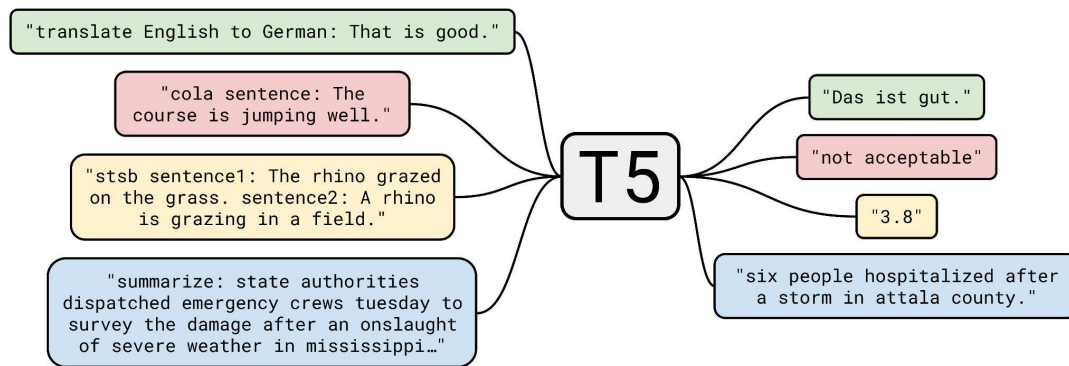
example : image retrieval



How to learn from Multimodality ?

Popular architecture : **Foundation model**

Large Language Model (LLM) : have a huge knowledge of the world and capacity of reasoning due to the quantity of data it saw during training, ex : BERT, GPT, T5, Palm, ...



How can I cook some cookies ?



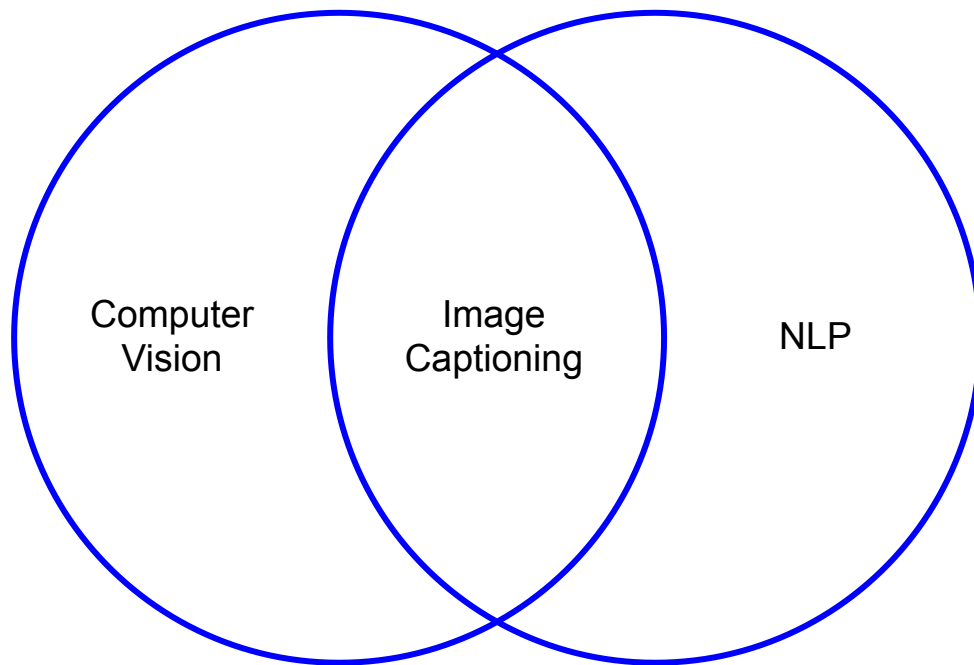
step 1 : ...
step 2 : ... (actions)

Multimodal Tasks

Multimodal Tasks - Image Captioning

Image Captioning is the process of generating textual description of an image

- Input : image
- Output : text



Multimodal Tasks - Image Captioning



Captioning Model

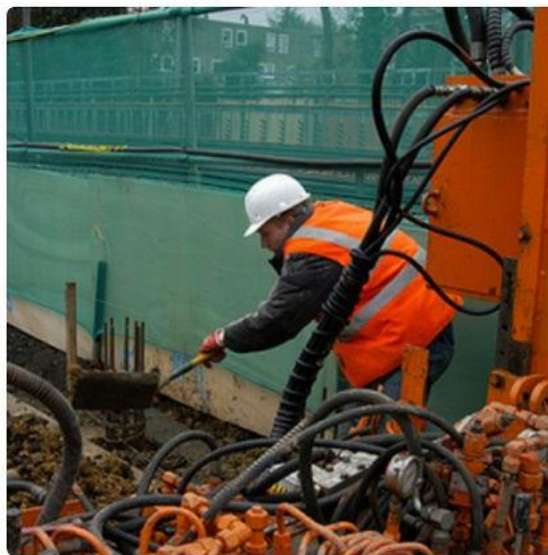
A happy dog is standing in the ocean

Multimodal Tasks - Image Captioning

Need textual and visual data for training



"man in black shirt is playing guitar."



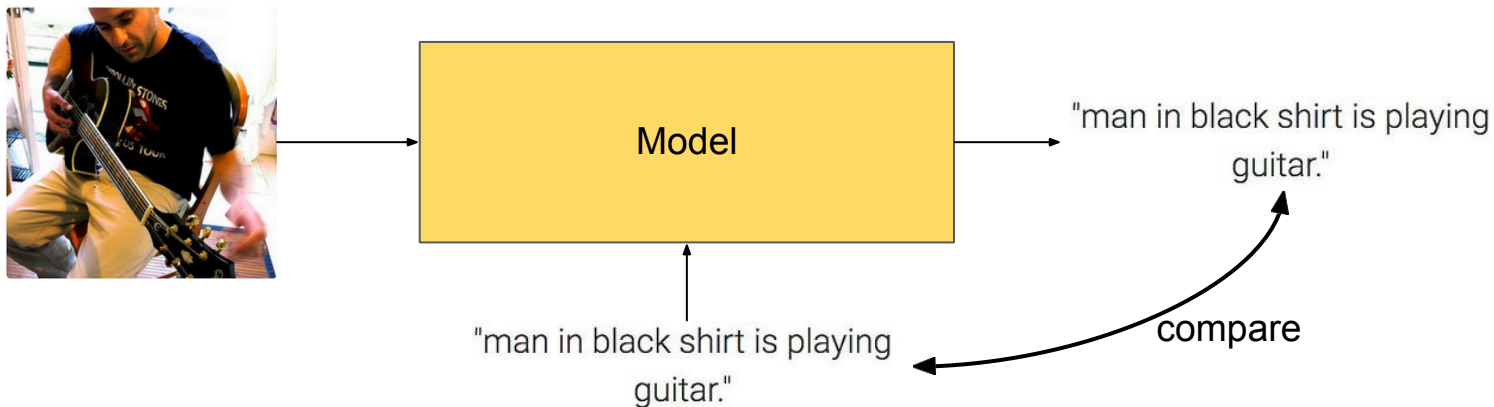
"construction worker in orange safety vest is working on road."



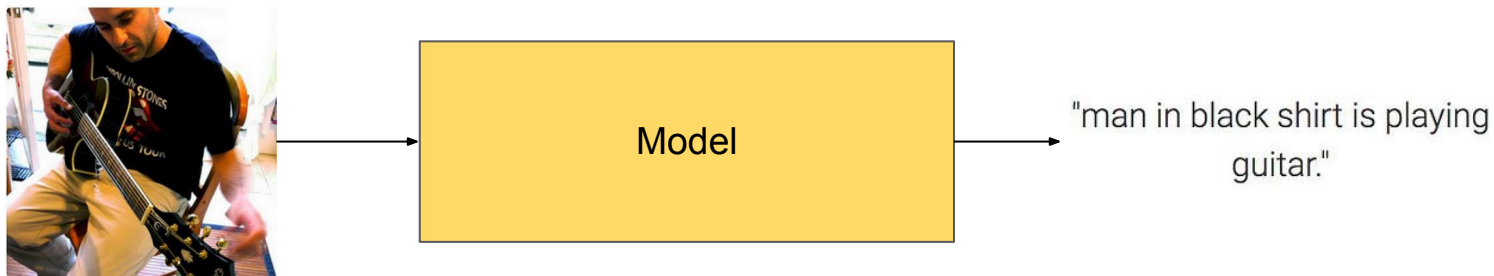
"two young girls are playing with lego toy."

Multimodal Tasks - Image Captioning

training :

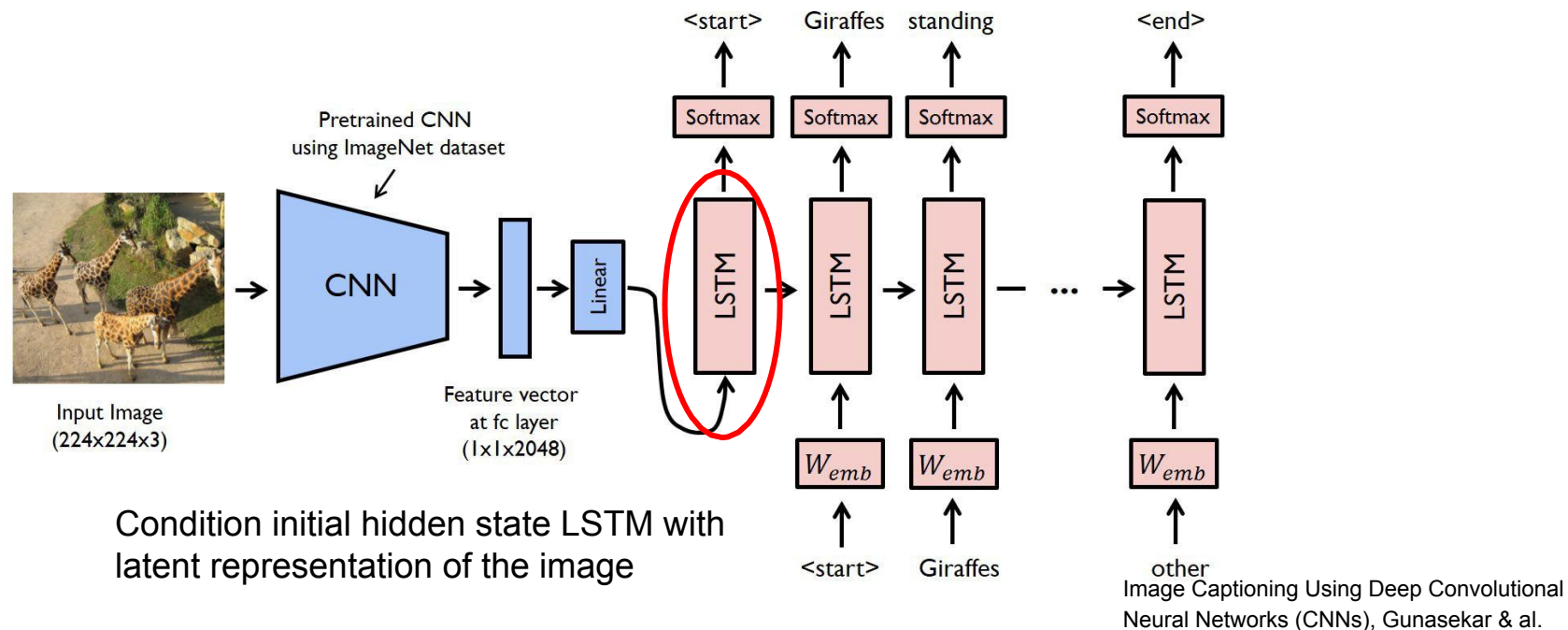


inference :



Multimodal Tasks - Image Captioning

Example of old architecture :



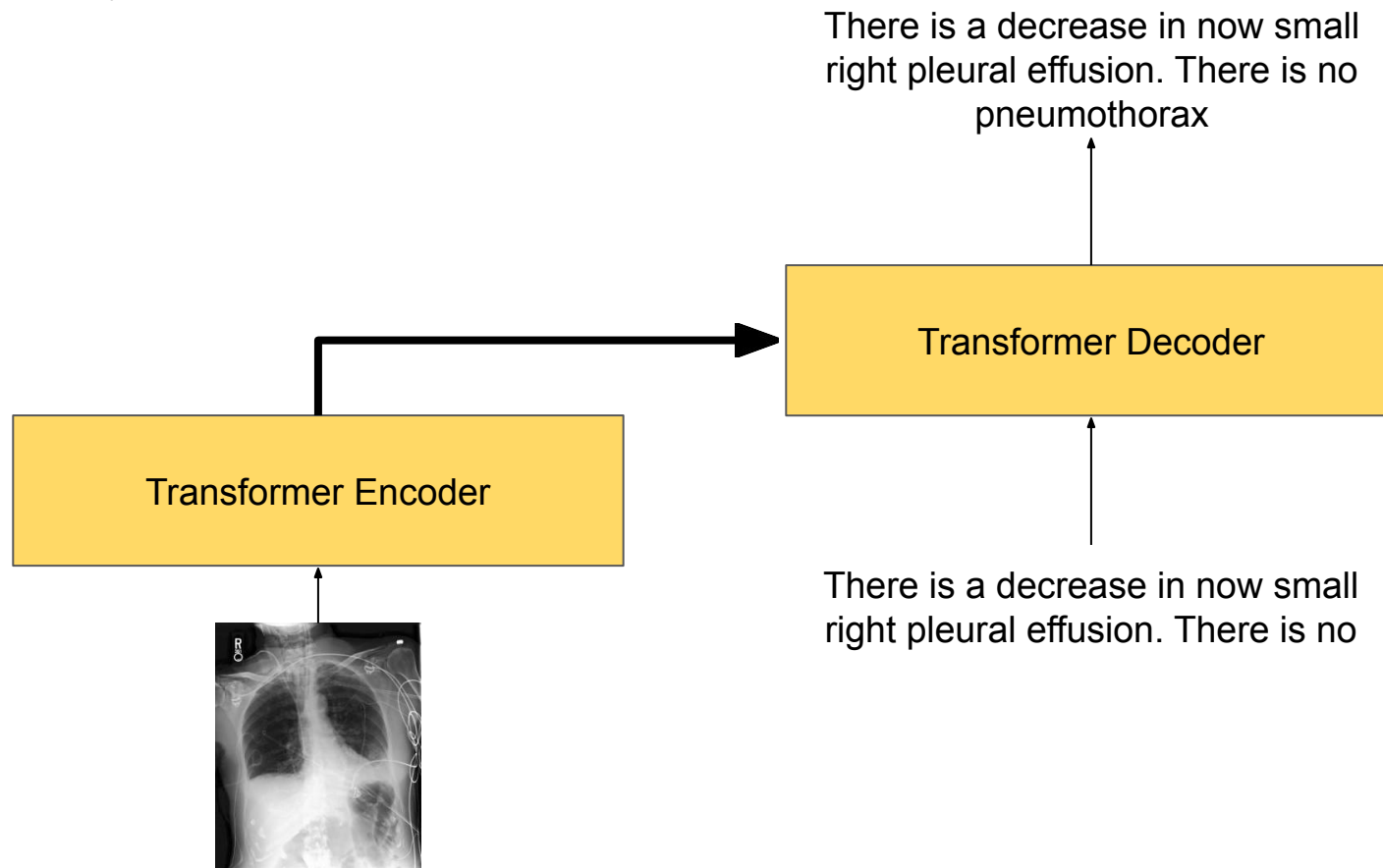
Turn each architecture to good account by using CNN for images (spatial data) and LSTM for text (sequential data)

Multimodal Tasks - Image Captioning

Chest X-Ray	Ground Truth	Our predictions
	Lungs remain well inflated without evidence of focal airspace consolidation, pleural effusions, pulmonary edema or pneumothorax. Irregularity in the right humeral neck is related to a known healing fracture secondary to recent fall. PA and lateral views of the chest _____ at 09:55 are submitted.	no findings. no pneumonia. no pleural effusion. no edema. there is little change and no evidence of acute cardiopulmonary disease. no pneumonia, vascular congestion, pleural effusion. of incidental note is an azygos fissure, of no clinical significance. this raises possibility of a normal variant.
	1. Stable bilateral small pleural effusions and atelectasis. 2. Enlarged pulmonary artery, suggesting pulmonary hypertension. No significant interval change. Bilateral small pleural effusions and adjacent atelectasis are overall unchanged. The heart is top-normal in size, unchanged. The pulmonary artery is enlarged, suggesting pulmonary hypertension. No demand, focal consolidation to suggest pneumonia, or pneumothorax.	pleural effusion present. lung opacity present. no edema. cardiomegaly present. atelectasis present. as compared to previous radiograph, there is an increase in extent of a pre existing small left pleural effusion with subsequent atelectasis at left lung bases. otherwise, radiograph is unchanged. moderate cardiomegaly. mild fluid overload no overt pulmonary edema. no new focal parenchymal opacities suggesting pneumonia. unchanged position of right pectoral port a cath.
	There is decrease in now small right pleural effusion. There is no pneumothorax. There is a new right pacer pigtail catheter. Cardiomedastinal contours are unchanged. Lines and tubes are in standard position. Left lower lobe opacities, a combination of pleural effusion and atelectasis, are unchanged.	uncertain pneumonia. pleural effusion present. lung opacity present. atelectasis present. bilateral pleural effusions, left greater than right. bibasilar opacities potentially atelectasis in setting of low lung volumes. infection be excluded. frontal and lateral views of chest demonstrate low lung volumes, which accentuate bronchovascular markings. there are small bilateral pleural effusions, right greater than left, with adjacent atelectasis. there is no focal consolidation pneumothorax. cardiomedastinal silhouette is within normal limits. surgical clips are seen in right upper quadrant of abdomen. aortic arch calcifications are noted.

Multimodal Tasks - Image Captioning

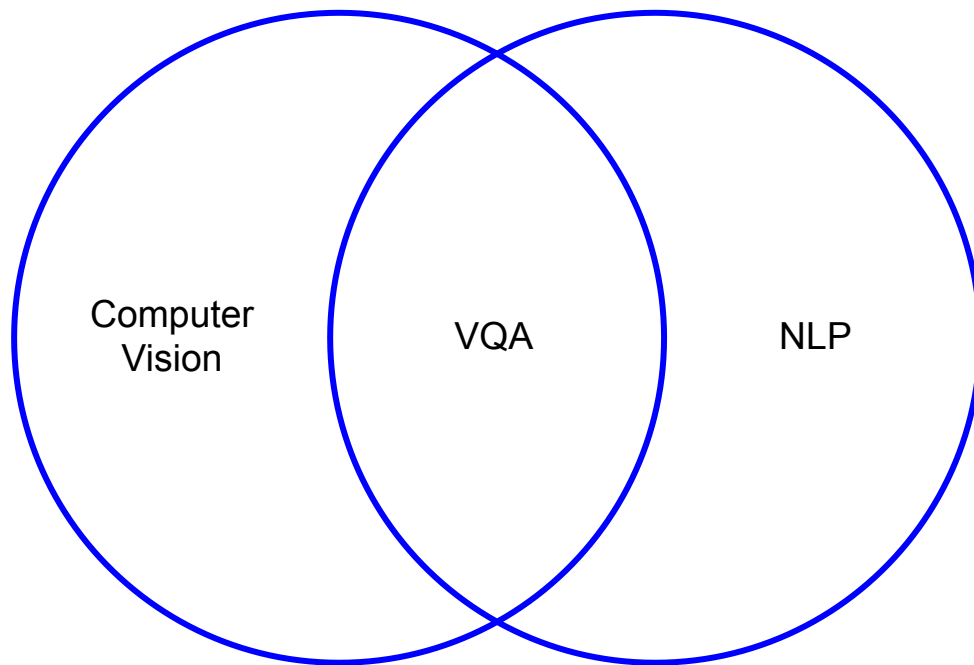
Architecture mostly used : **Transformers**



Multimodal Tasks - VQA

Visual Question Answering is the task of answering open-ended questions based on an image

- Input : text + image
- Output : text



Multimodal Tasks - VQA



Multimodal Tasks - VQA

Need textual and visual data for training

Que: What kind of plants are these?

Ans: flowers



Que: Is the kite high in the air?

Ans: yes



Que: Where is the light coming from?

Ans: window

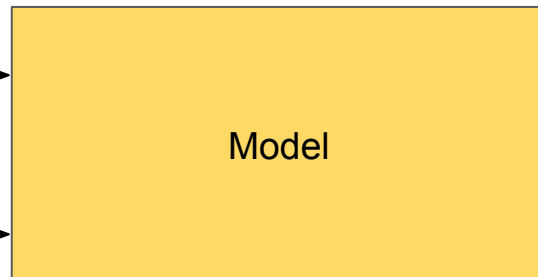


Multimodal Tasks - Image Captioning

training :



Que: What kind of plants are these?



Ans: flowers

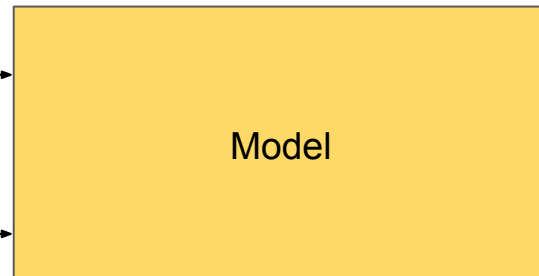
flowers

compare

inference :



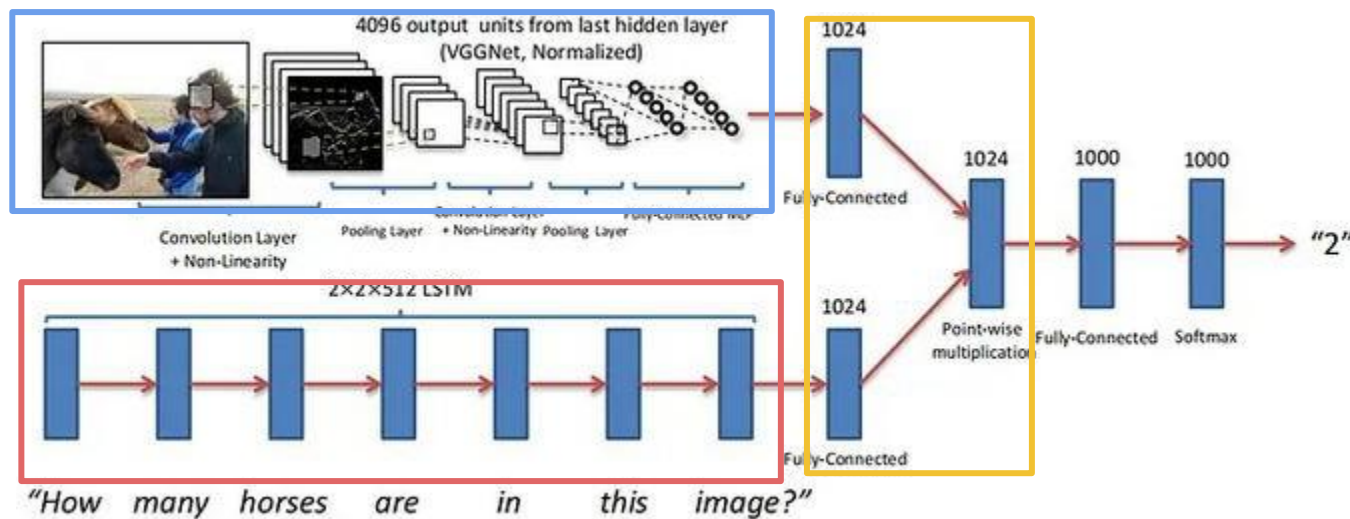
Que: What kind of plants are these?



Ans: flowers

Multimodal Tasks - VQA

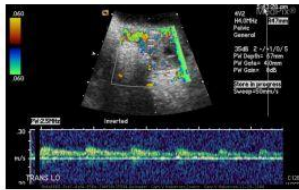
Example of old architecture :



Fusion approach to treat multimodal data : Specialized architecture to encode **text** and **images** with linear layers and pointwise multiplication to **merge** the latent representations

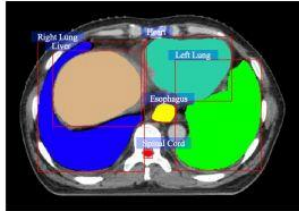
Multimodal Tasks - VQA

VQA-Med-2020 [13]



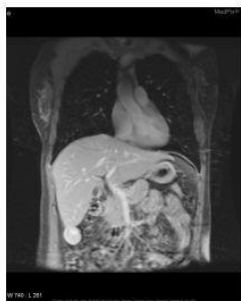
Q: what abnormality is seen in the image?
A: ovarian torsion

SLAKE [57]

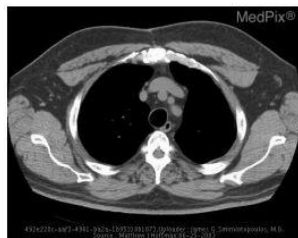


Q: Does the image contain left lung?
A: Yes

VQA-Med-2021 [15]



Q: What is most alarming about this mri?
A: focal nodular hyperplasia



Q: what is abnormal in the ct scan?
A: partial anomalous pulmonary venous return

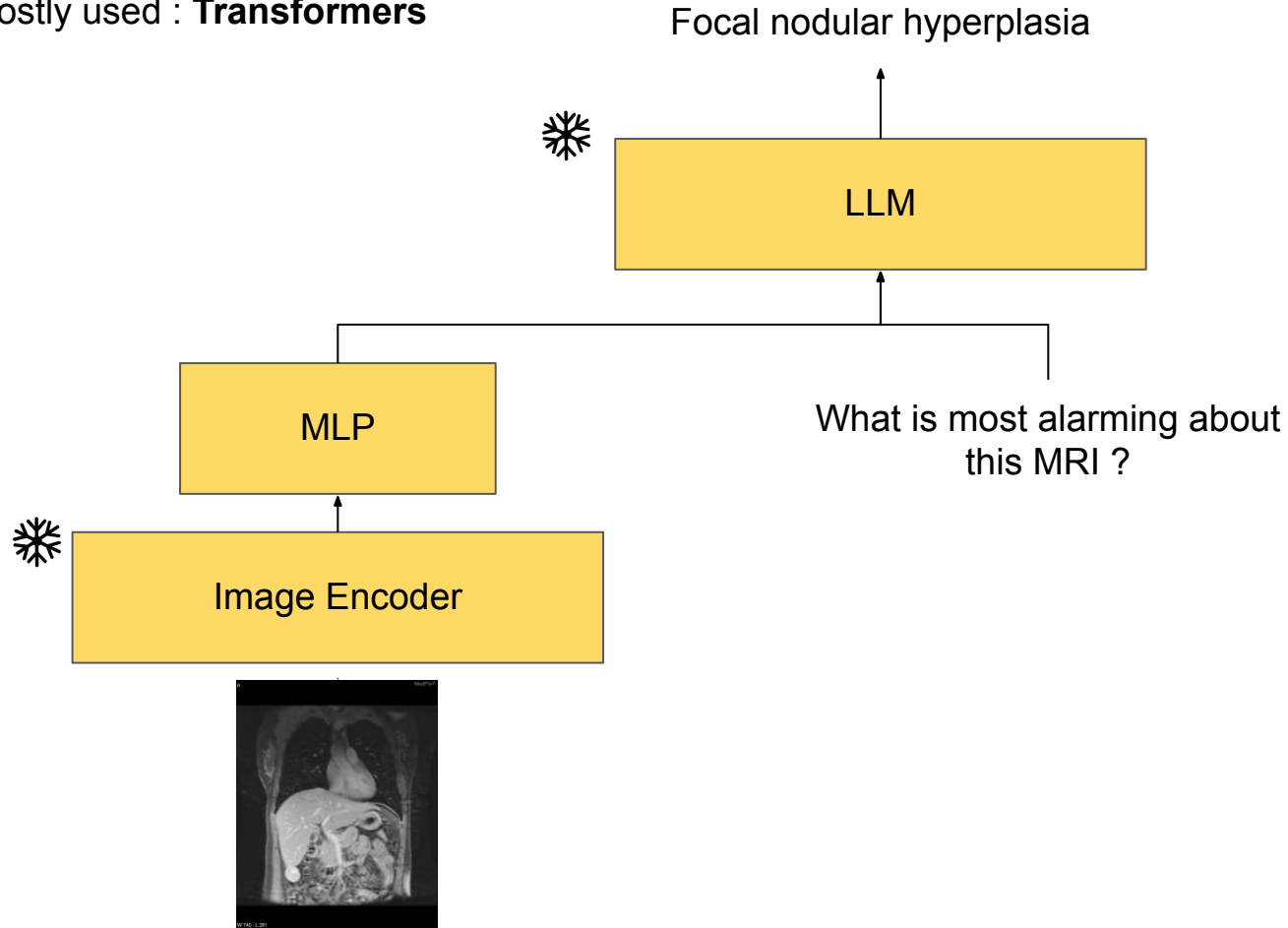
Q: What is the function of the rightmost organ in this picture?
A: Breathe



Q: What abnormality is seen in the image?
A: Enhancing lesion right parietal lobe with surrounding edema

Multimodal Tasks - VQA

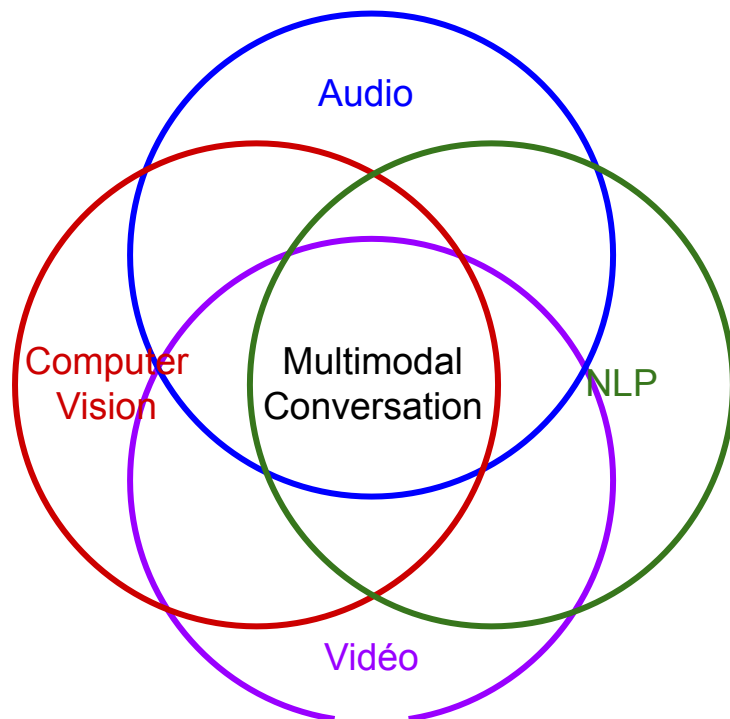
Architecture mostly used : **Transformers**



Multimodal Tasks - Multimodal Conversational System

A **multimodal conversational AI system** answers questions, fulfills tasks, and emulates human conversations by understanding and expressing itself via multiple modalities.

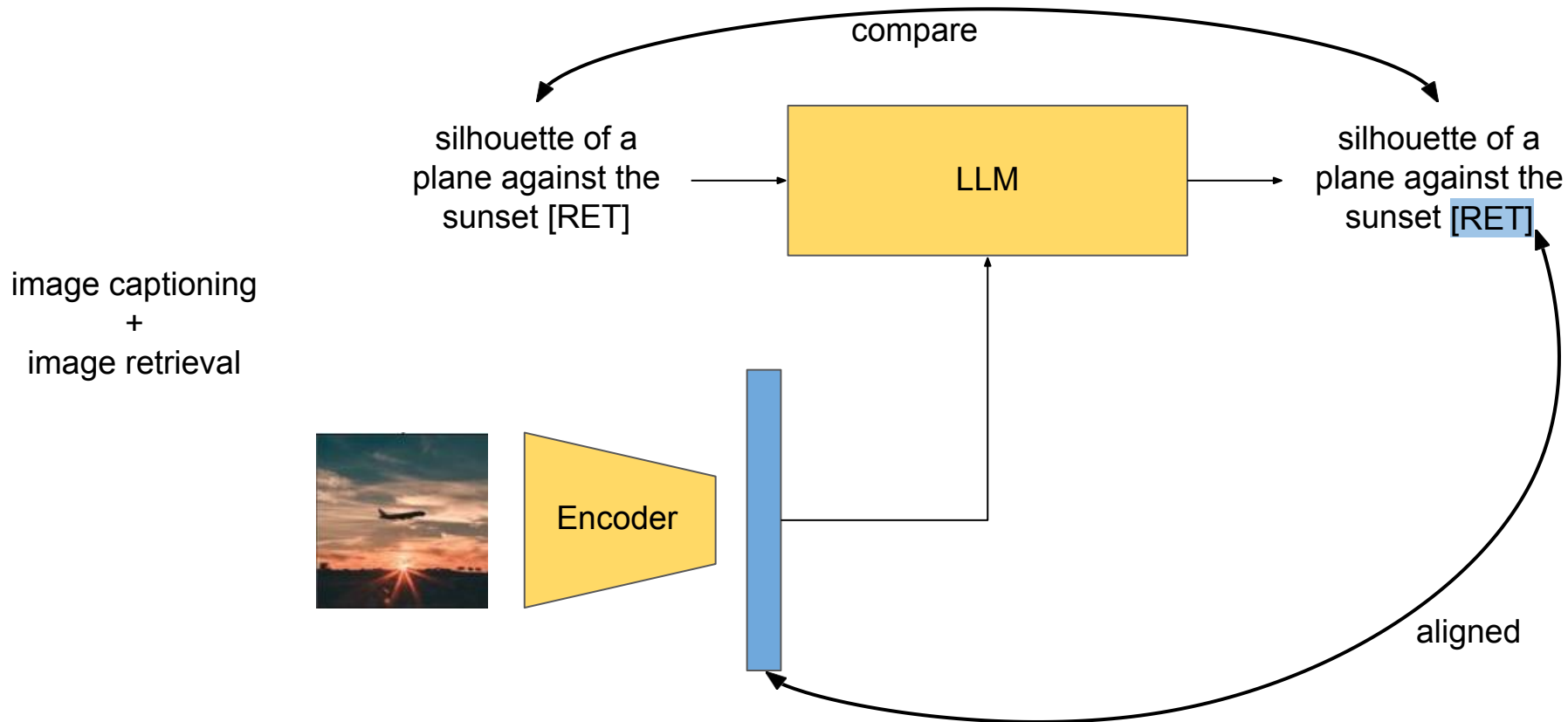
- Input : text, image, video, audio, ...
- Output : text, image, video, audio, ...



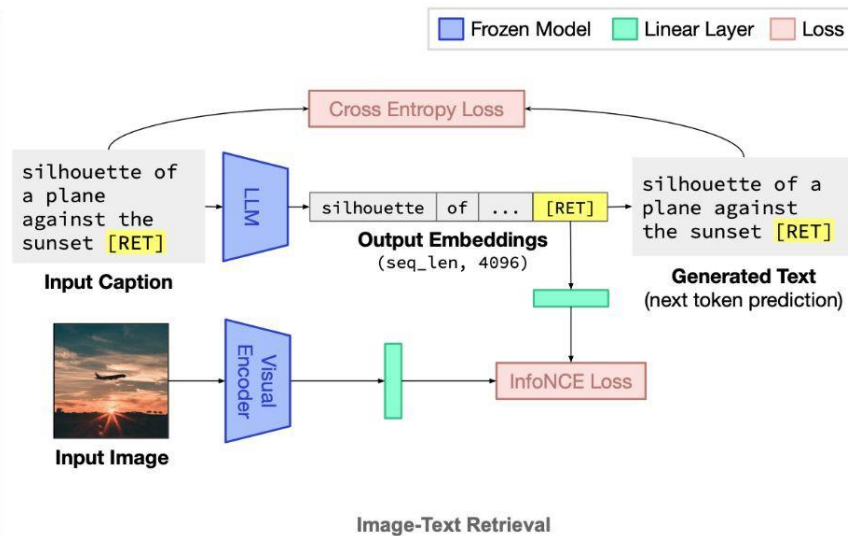
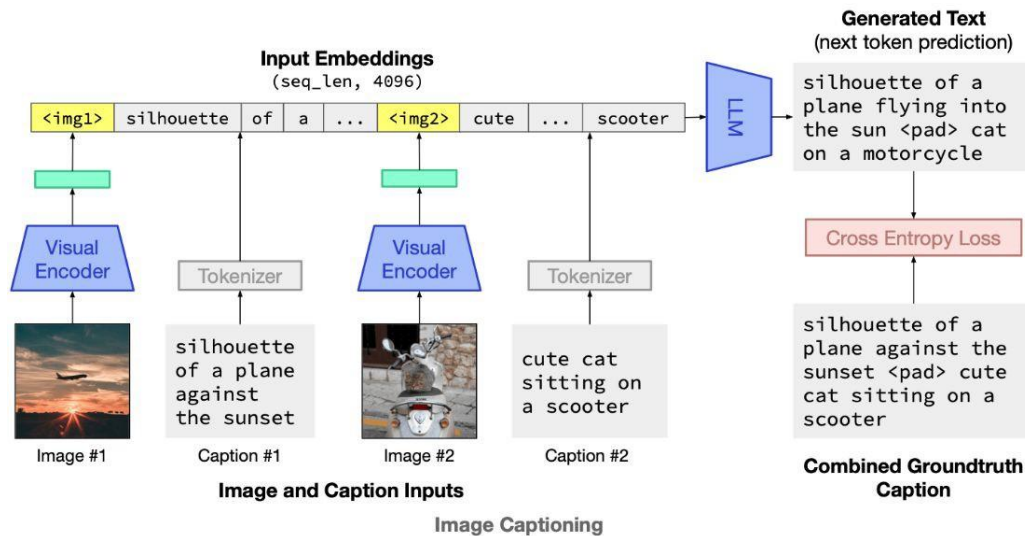
Multimodal Tasks - Multimodal Conversational System



Multimodal Tasks - Multimodal Conversational System



Multimodal Tasks - Multimodal Conversational System



Grounding Language Models to Images for Multimodal Inputs and Outputs;
Koh, Jing Yu; Salakhutdinov; Ruslan and Fried, Daniel

Multimodal Tasks - Multimodal Conversational System



Instructions: You are a helpful radiology assistant. Describe what lines, tubes, and devices are present and each of their locations. Describe if pneumothorax is present; if present, describe size on each side. Describe if pleural effusion is present; if present, describe amount on each side. Describe if lung opacity (atelectasis, fibrosis, consolidation, infiltrate, lung mass, pneumonia, pulmonary edema) is present; if present, describe kinds and locations. Describe the cardiac silhouette size. Describe the width and contours of the mediastinum. Describe if hilar enlargement is present; if enlarged, describe side. Describe what fractures or other skeletal abnormalities are present.

Given the LATERAL view X-ray image . **Reason for the study:** Amiodarone routine surveillance.

Q: Describe the findings in the image following the instructions.

A: As compared to the previous radiograph there is no relevant change. Normal lung volumes. Mild bilateral apical scarring. Normal size of the cardiac silhouette and tortuosity of the thoracic aorta. No pathologic findings in the lung parenchyma notably no evidence of fibrotic lung parenchymal changes. A faint 2 mm rounded opacity projecting over the lower aspect of the fourth right rib and internally to the upper border of the second right rib is seen on the frontal radiograph only and likely reflects structure on the skin.

Given the PA view X-ray image . **Reason for the study:** History m with malaise pneumonia.

Q: Describe the findings in the image following the instructions.

A:



Instructions: You are a helpful dermatology assistant. The following are questions about skin lesions. Categorize the skin lesions into the most likely class given the patient history.

Given . **Patient History:** Age: 51, Gender: female, Smoke: false, Drink: false, Family skin cancer history: true, Family any cancer history: false, Lesion region: back, Lesion itch: false, Lesion grew: false, Lesion bled: false, Lesion elevation: false, Fitzpatrick scale: 1.0, Diameters (mm): [12.0, 8.0].

Q: Which of the following is the most likely diagnosis of the patient's skin lesion?

(A) Nevus (B) Basal Cell Carcinoma (C) Squamous Cell Carcinoma (D) Actinic Keratosis (E) Seborrheic Keratosis (F) Melanoma

A: Basal Cell Carcinoma.

Given . **Patient History:** Age: 39, Gender: unknown, Smoke: unknown, Drink: unknown, Family skin cancer history: unknown, Family any cancer history: unknown, Lesion region: neck, Lesion itch: false, Lesion grew: true, Lesion bled: false, Lesion elevation: true, Fitzpatrick scale: unknown, Diameters (mm): [unknown, unknown].

Q: Which of the following is the most likely diagnosis of the patient's skin lesion?

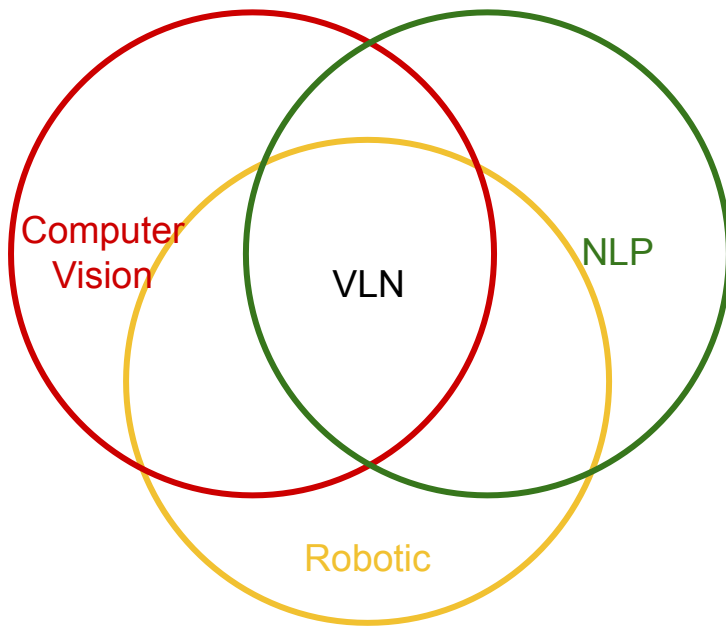
(A) Nevus (B) Basal Cell Carcinoma (C) Squamous Cell Carcinoma (D) Actinic Keratosis (E) Seborrheic Keratosis (F) Melanoma

A:

Multimodal Tasks

Vision-and-Language Navigation aims to build an embodied agent that can communicate with humans in natural language and navigate in real 3D environments

- Input : text, image
- Output : text, image, actions



Multimodal Tasks



Pass the pool and go indoors using the double glass doors. Pass the large table with chairs and turn left and wait by the wine bottles that have grapes by them.

Walk straight through the room and exit out the door on the left. Keep going past the large table and turn left. Walk down the hallway and stop when you reach the 2 entry ways. One in front of you and one to your right. The bar area is to your left.

Enter house through double doors, continue straight across dining room, turn left into bar and stop on the circle on the ground.



Standing in front of the family picture, turn left and walk straight through the bathroom past the tub and mirrors. Go through the doorway and stop when the door to the bathroom is on your right and the door to the closet is to your left.

Walk with the family photo on your right. Continue straight into the bathroom. Walk past the bathtub. Stop in the hall between the bathroom and toilet doorways.

Walk straight passed bathtub and stop with closet on the left and toilet on the right.



Exit the office then turn left and then turn left in the hallway and head down the hallway until you get to a door on your left and go into office 359 then stop.

Go out of the room and take a left. Go into the first room on your left.

Leave the office and take a left. Take the next left at the hallway. Walk down the hall and enter the first office on the left. Stop next to the door to office 359.

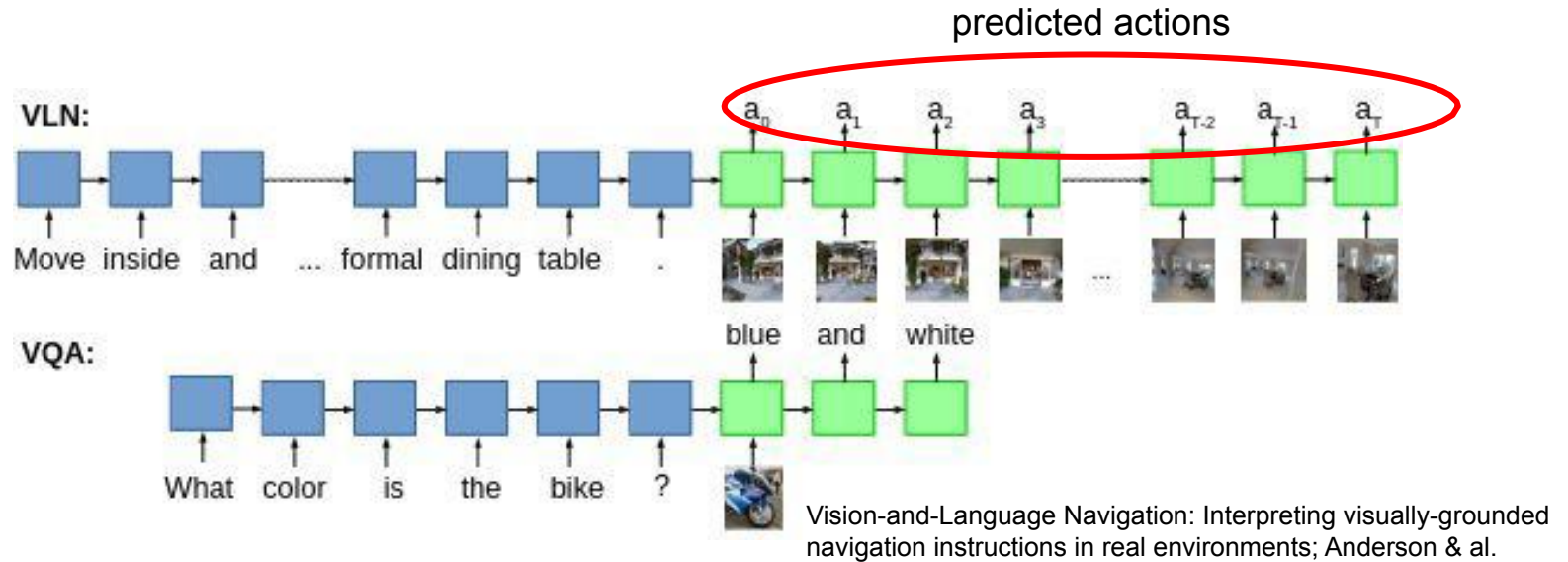


Go up the stairs and turn right. Go past the bathroom and stop next to the bed.

Walk all the way up the stairs, and immediately turn right. Pass the bathroom on the left, and enter the bedroom that is right there, and stop there.

Walk up the stairs turn right at the top and walk through the doorway continue straight and stop inside the bedroom.

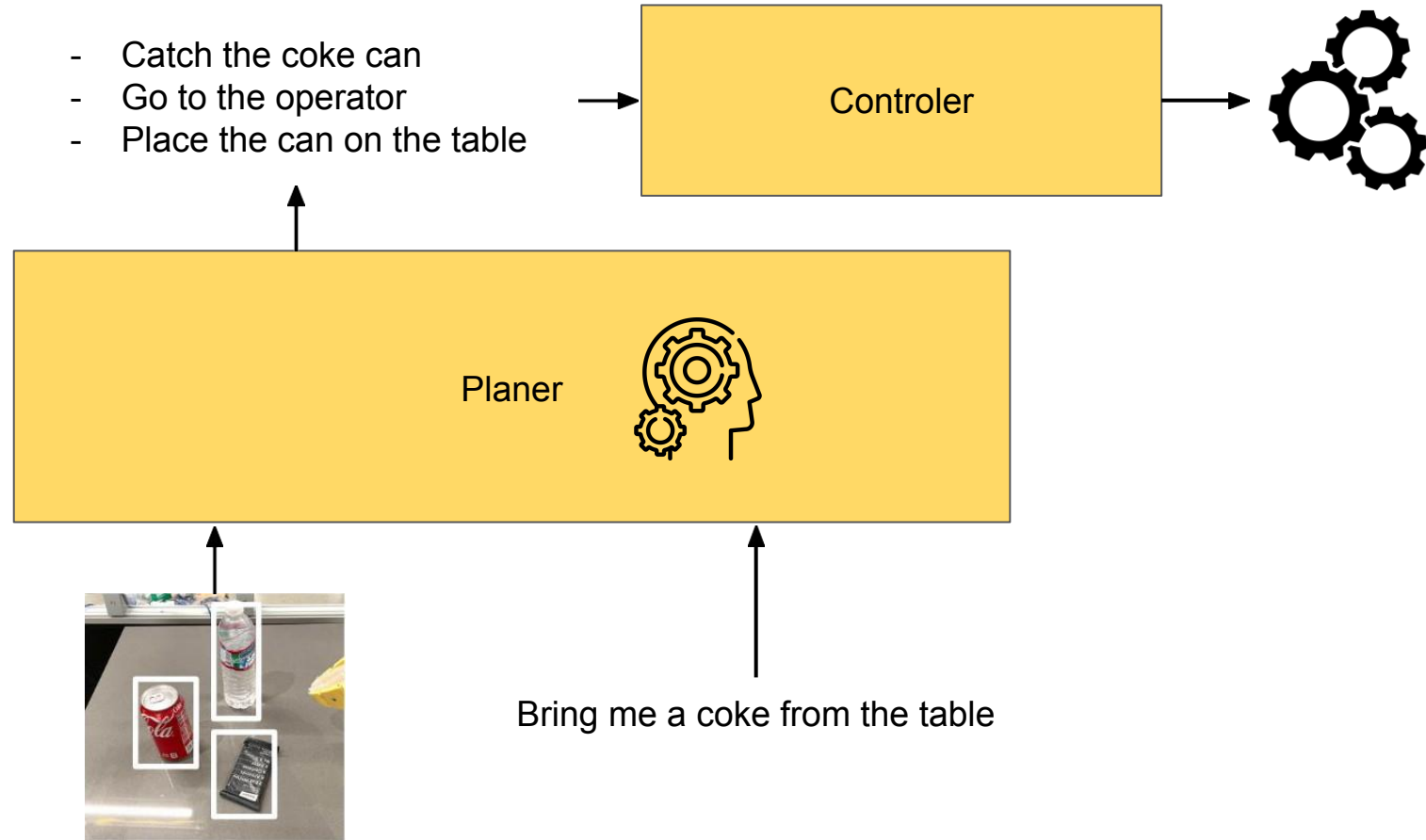
Multimodal Tasks



Use LSTM for text $\rightarrow$ Normal

Use LSTM for images $\rightarrow$????? : Normal because images are correlated in time for this case : it's like a video

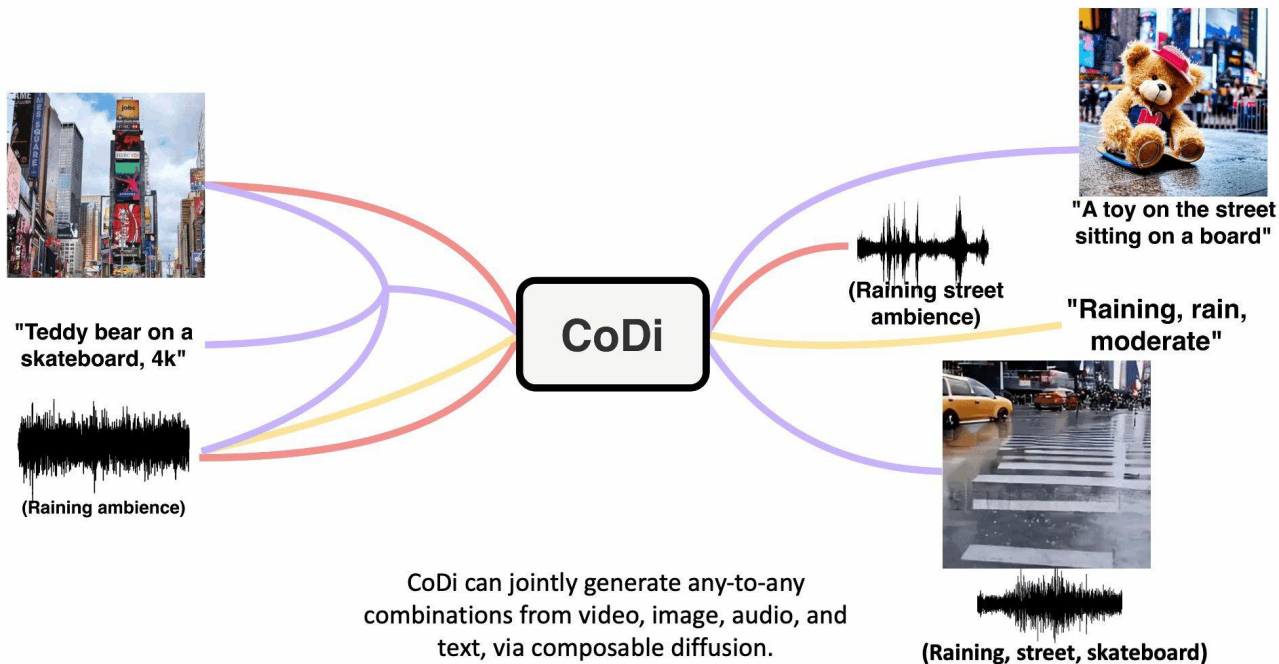
Multimodal Tasks



Examples of Models

Examples of Models

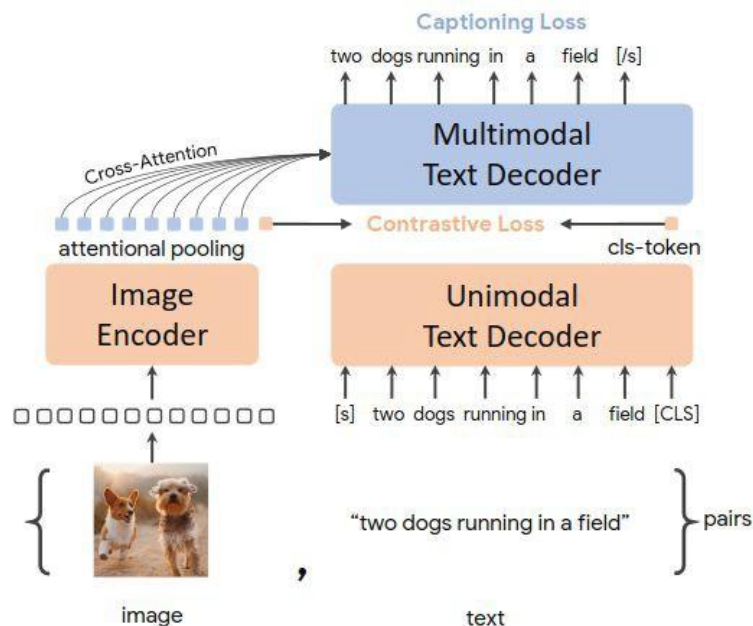
CoDi : Any to any generation via composable diffusion



idea : aligned all modalities in a latent space and use it to condition diffusion models (one for each modality)

Examples of Models

CoCa : Learn a contrastive captioner to be a image-text foundation model



Algorithm 1 Pseudocode of Contrastive Captioners architecture.

```
# image, text.ids, text.labels, text.mask: paired {image, text} data
# con_query: 1 query token for contrastive embedding
# cap_query: N query tokens for captioning embedding
# cls_token_id: a special cls_token_id in vocabulary

def attentional_pooling(features, query):
    out = multihead_attention(features, query)
    return layer_norm(out)

img_feature = vit_encoder(image) # [batch, seq_len, dim]
con_feature = attentional_pooling(img_feature, con_query) # [batch, 1, dim]
cap_feature = attentional_pooling(img_feature, cap_query) # [batch, N, dim]

ids = concat(text.ids, cls_token_id)
mask = concat(text.mask, zeros_like(cls_token_id)) # unpad cls_token_id
txt_embs = embedding_lookup(ids)
unimodal_out = lm_transformers(txt_embs, mask, cross_attn=None)
multimodal_out = lm_transformers(
    unimodal_out[:, :-1, :], mask, cross_attn=cap_feature)
cls_token_feature = layer_norm(unimodal_out)[:, -1:, :] # [batch, 1, dim]

con_loss = contrastive_loss(con_feature, cls_token_feature)
cap_loss = softmax_cross_entropy_loss(
    multimodal_out, labels=text.labels, mask=text.mask)
```

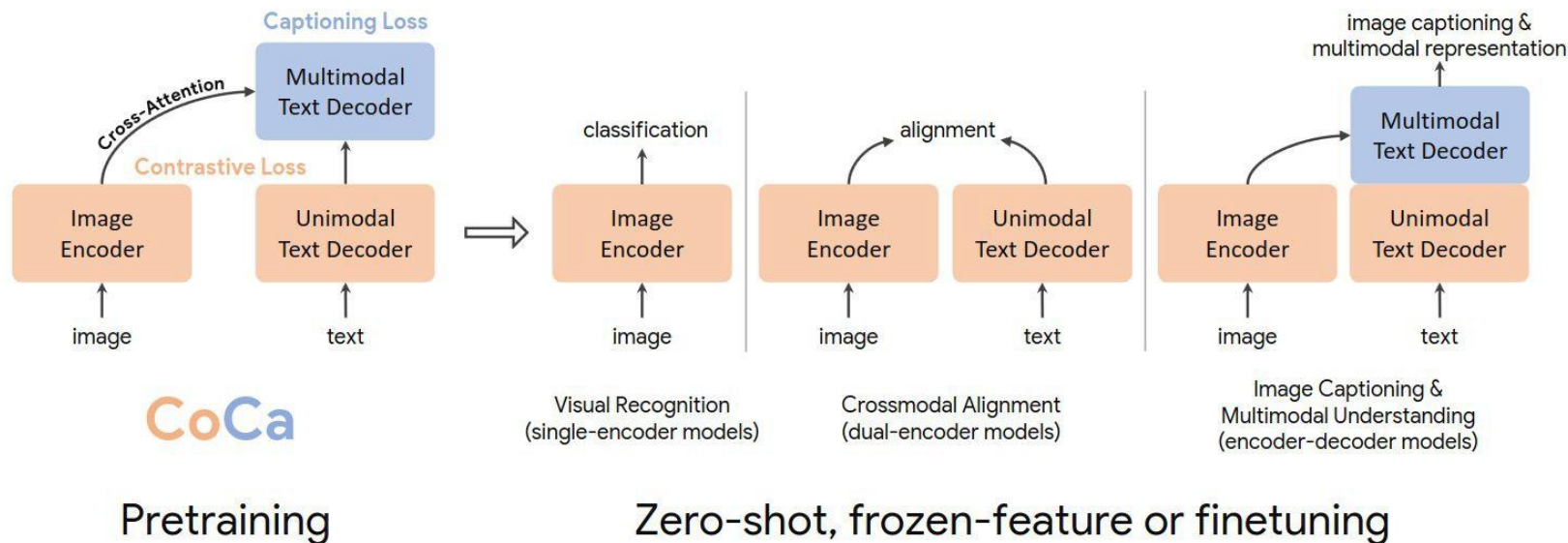
vit_encoder: vision transformer based encoder; *lm_transformer*: language-model transformers.

Figure 2: Detailed illustration of CoCa architecture and training objectives.

idea : Build a Foundation Model that do captioning and image-text alignment so that it keeps relevant information in the latent space.

Examples of Models

CoCa : Learn a contrastive captioner to be a image-text foundation model

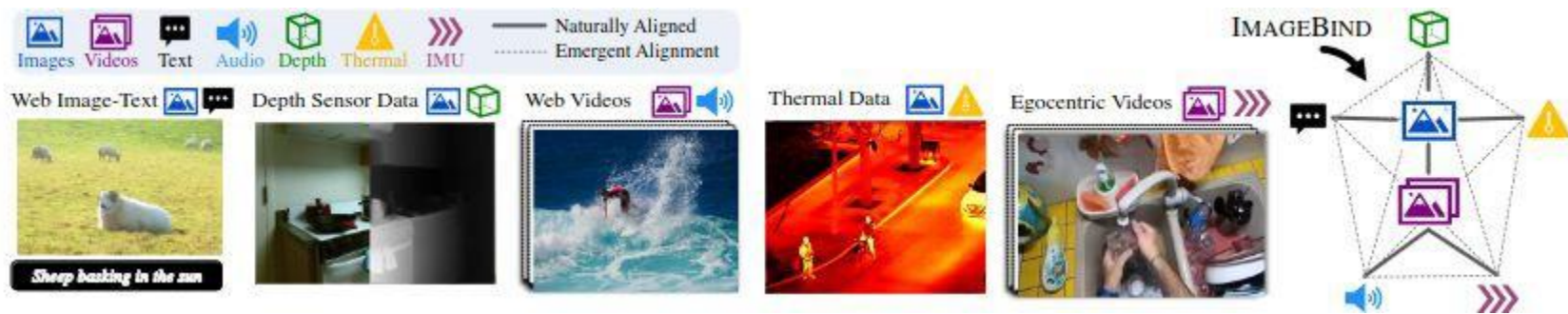


The goal is to use CoCa on :

- zero-shot : use CoCa on unseen tasks without adding any training
- frozen-feature : use CoCa on seen tasks but with unseen dataset
- finetuning : train a bit CoCa on a task or data that it has never seen

Examples of Models

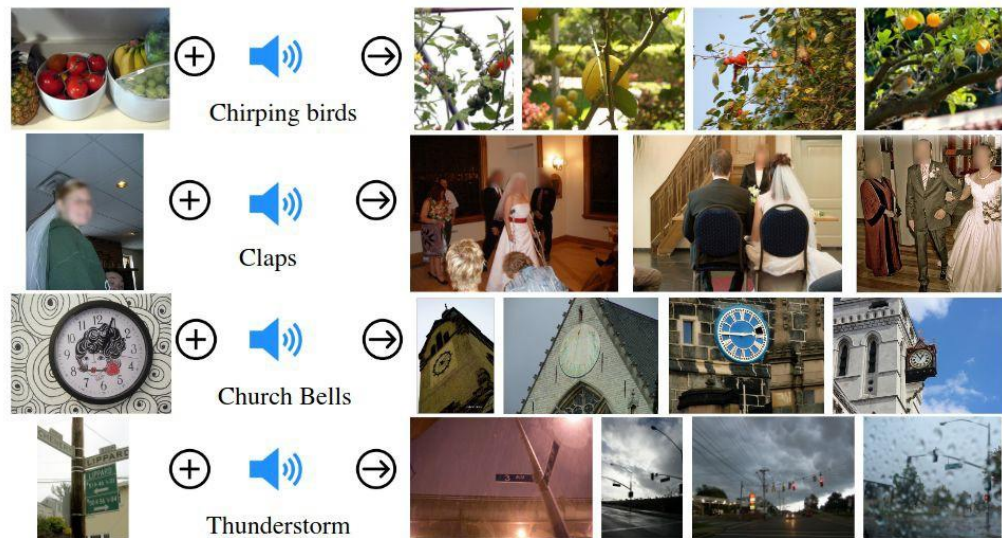
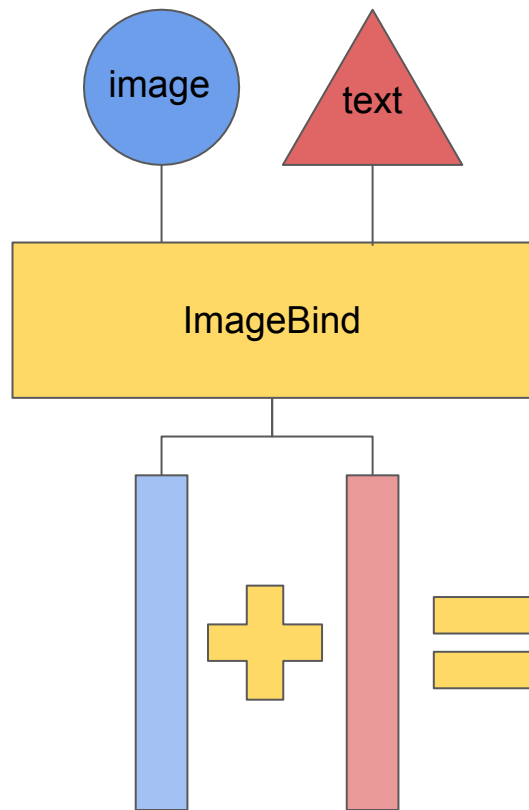
ImageBind : Multimodal alignment using transformer encoders



idea : use naturally aligned modalities (image-video, image-text, image-depth, video-audio, ...) to align all the modalities

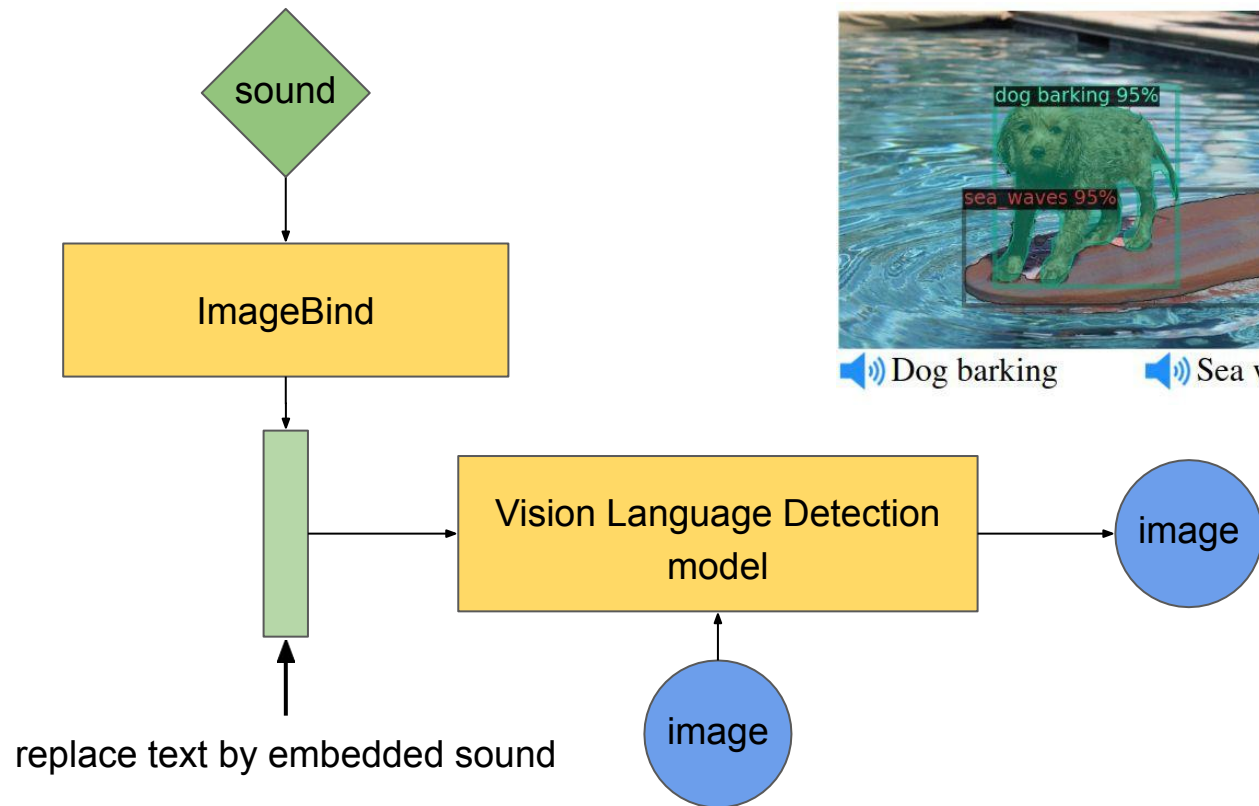
Examples of Models

ImageBind : Multimodal alignment using transformer encoders



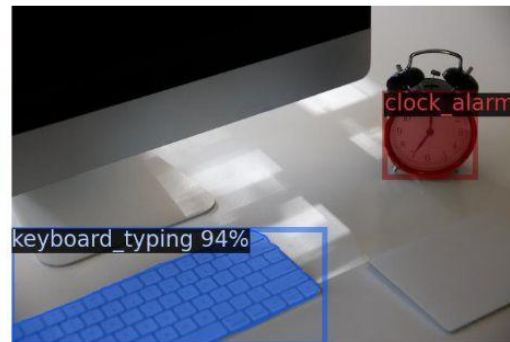
Examples of Models

ImageBind : Multimodal alignment using transformer encoders



🔊 Dog barking

🔊 Sea waves



🔊 Keyboard typing

🔊 Clock alarm

Examples of Models

BLIP-2 : Vision-Language Model using pre-trained models



Explain the advantages of this product.

The audi e-tron quattro concept is a plug-in hybrid electric sports car that has a range of 310 miles and can accelerate from 0 to 60 mph in just 4 seconds.



Tell me something about the history of this place.

The great wall of china was built in 221 bc by emperor qin shi Huang to protect the emperor's capital from invaders from the north.



Write down the facts that you know about this flower.

It is a genus of flowering plants in the orchid family, orchidaceae. It is native to eastern asia and south east asia. It is a perennial flowering plant that grows up to a height of 1 m.



Is this photo unusual?

Yes, it's a house that looks like it's upside down.

How could someone get out of the house?

It has a slide on the side of the house.



What are shown in the photo?

A man and a chicken.

What does the man feel and why?

He is scared of the chicken because it is flying at him.



What are the ingredients I need to make this?

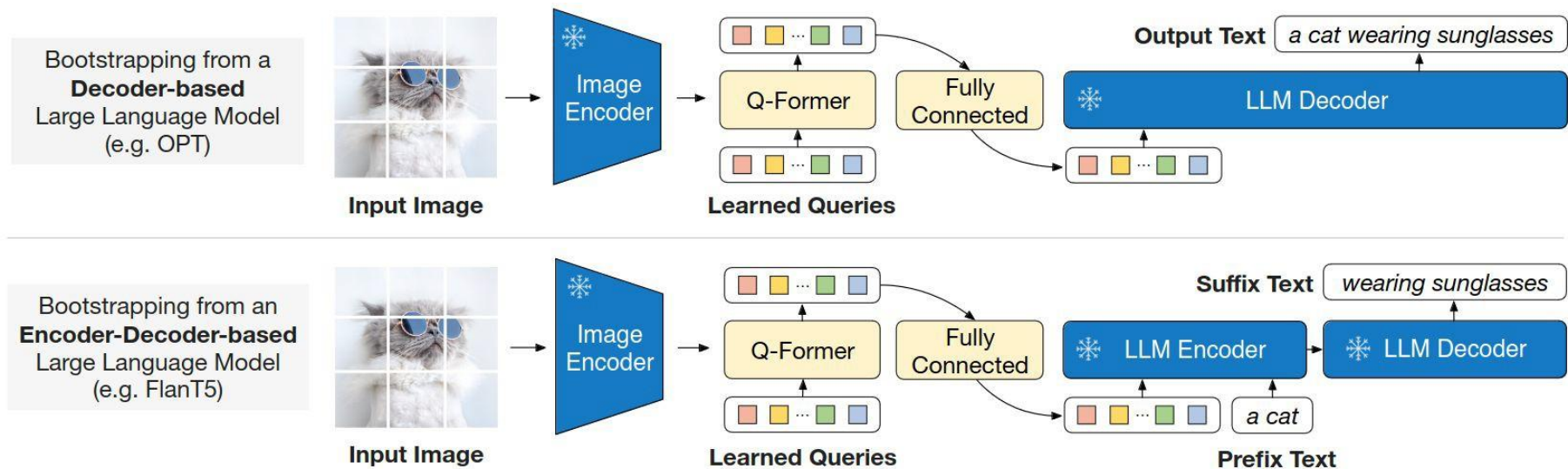
Pepperoni, mozzarella cheese, pizza sauce, olive oil, salt, pepper, basil.

What is the first step?

Place the pizza dough on a baking sheet, brush with olive oil, sprinkle with salt, pepper, and basil.

Examples of Models

BLIP-2 : Vision-Language Model using pre-trained models

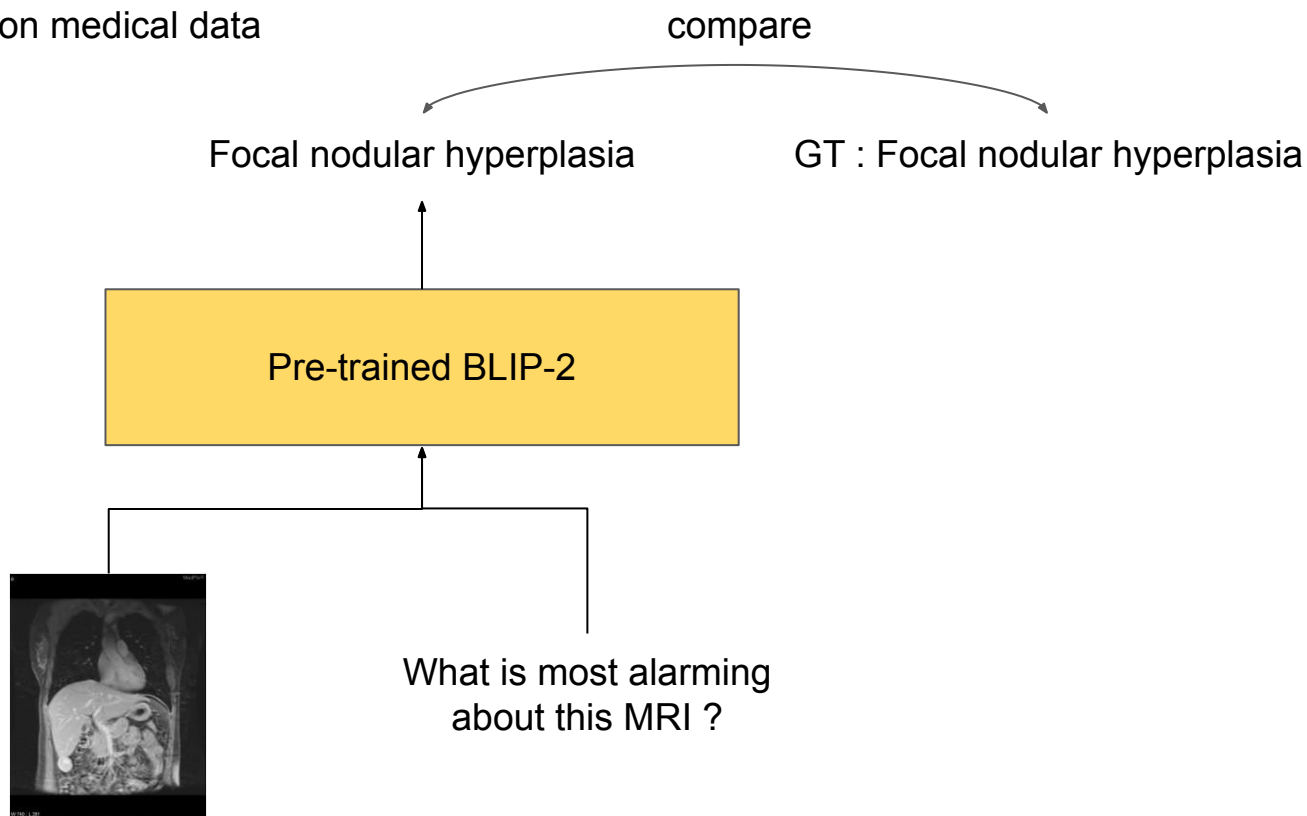


Q-Former : module train to bridge gap between frozen image encoder and a frozen LLM

Examples of Models

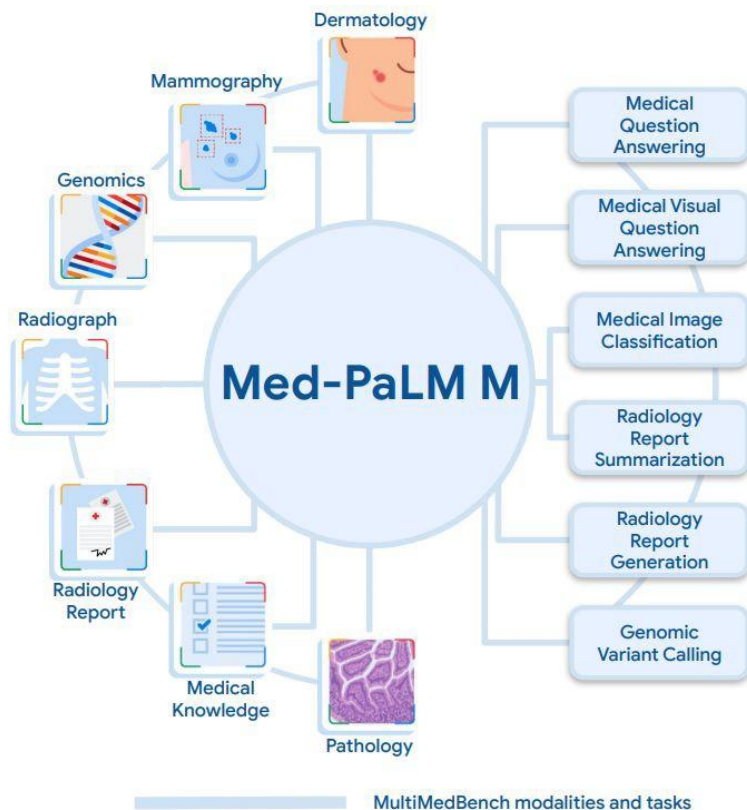
BLIP-2 : Vision-Language Model using pre-trained models

Fine-Tune the model on medical data
(done [here](#))



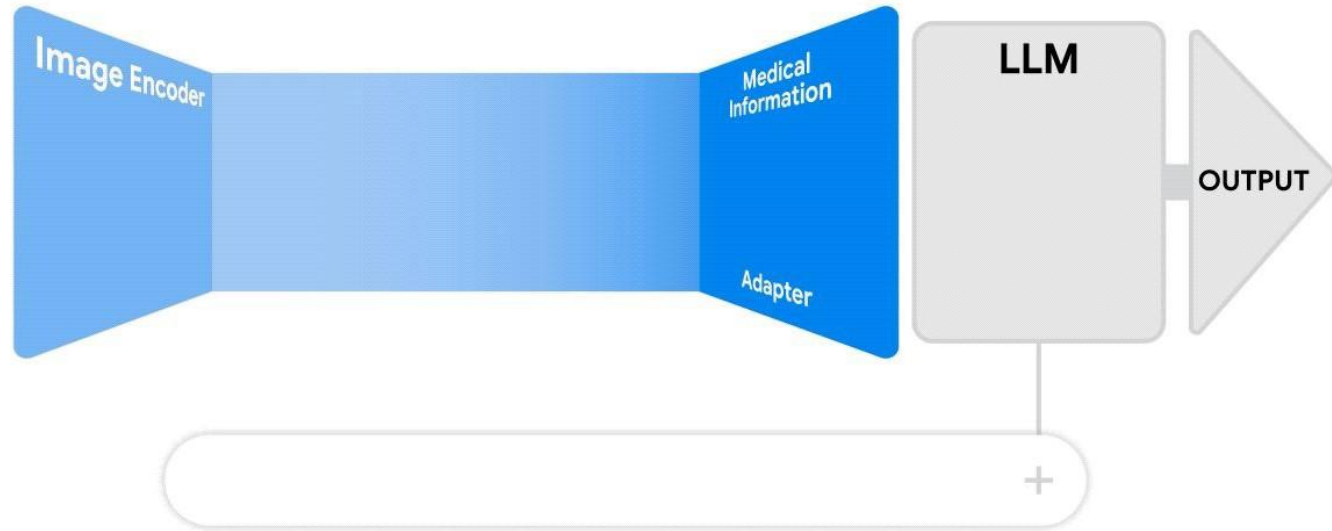
Examples of Models

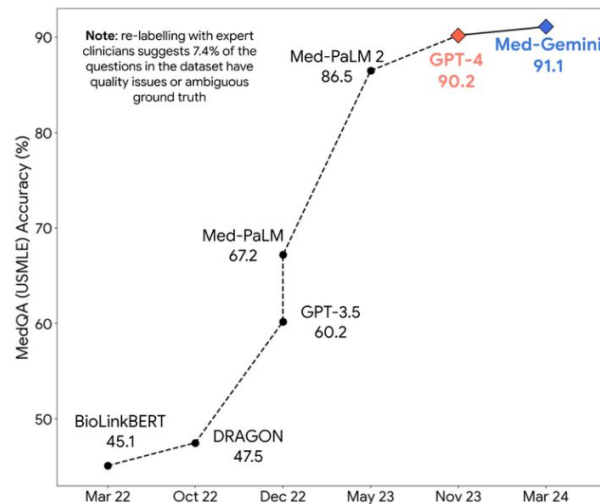
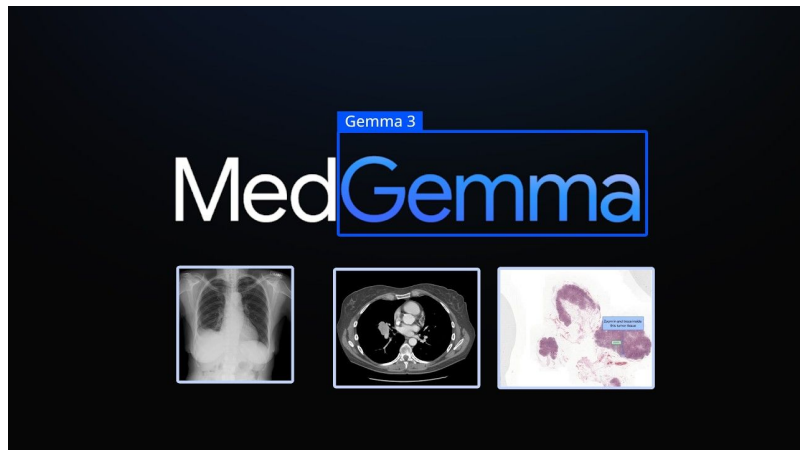
Med-PaLM : A Medical Large Language Model



Examples of Models

Med-PaLM : A Medical Large Language Model





[Open available](#), as it's a rather small model

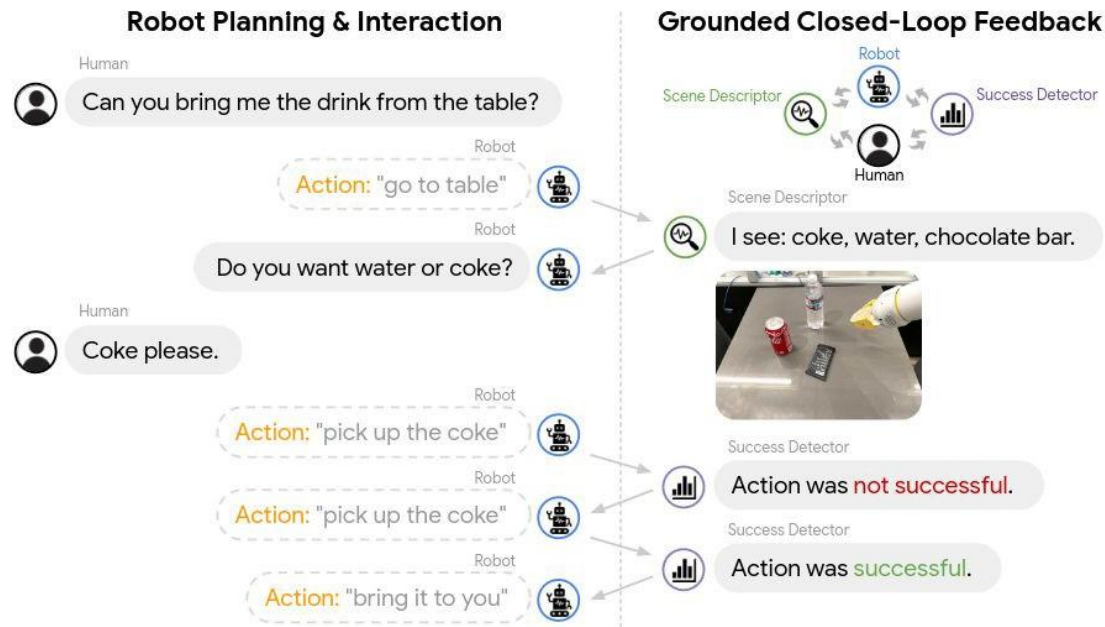


Metric	MedGemma 27B	Gemma 3 27B	MedGemma 4B	Gemma 3 4B
MedQA (4-op)	89.8 (best-of-5) 87.7 (0-shot)	74.9	64.4	50.7
MedMCQA	74.2	62.6	55.7	45.4
PubMedQA	76.8	73.4	73.4	68.4
MMLU Med (text only)	87.0	83.3	70.0	67.2
MedXpertQA (text only)	26.7	15.7	14.2	11.6
AfriMed-QA	84.0	72.0	52.0	48.0

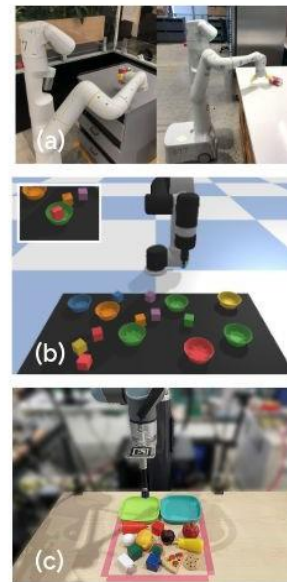
[Open available](#), as it's a rather small model

Examples of Models

InnerMonologue : Embodied Reasoning through Planning with Language Models



Robot Environments

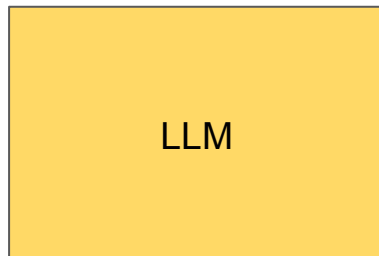


idea : use the knowledge of the LLM to plan a sequence of action and ground this LLM with modules that describe the scene with language

Examples of Models

InnerMonologue : Embodied Reasoning through Planning with Language Models

Go to the table

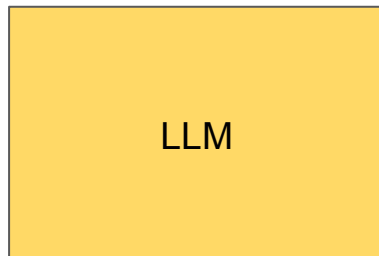


human : Bring me a drink from the table

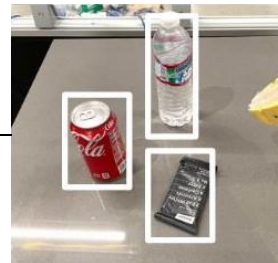
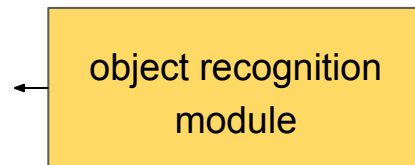
Examples of Models

InnerMonologue : Embodied Reasoning through Planning with Language Models

What kind of drink would you like ?



I see a coke and
lime soda



human : Bring me a drink from the table

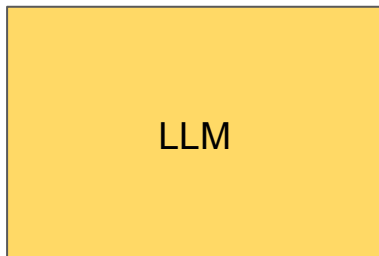
Robot : Go to the table

Scene : I see a coke and lime soda

Examples of Models

InnerMonologue : Embodied Reasoning through Planning with Language Models

Pick up coke



human : Bring me a drink from the table

Robot : Go to the table

Scene : I see a coke and lime soda

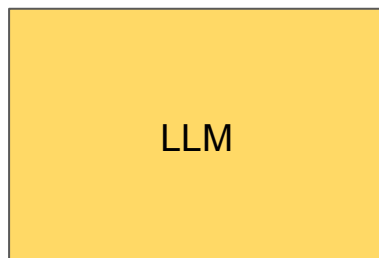
Robot : What kind of drink would you like ?

Human : Something with caffeine

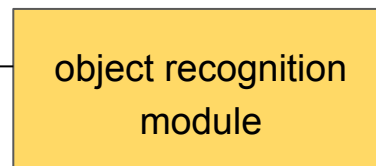
Examples of Models

InnerMonologue : Embodied Reasoning through Planning with Language Models

Pick up coke



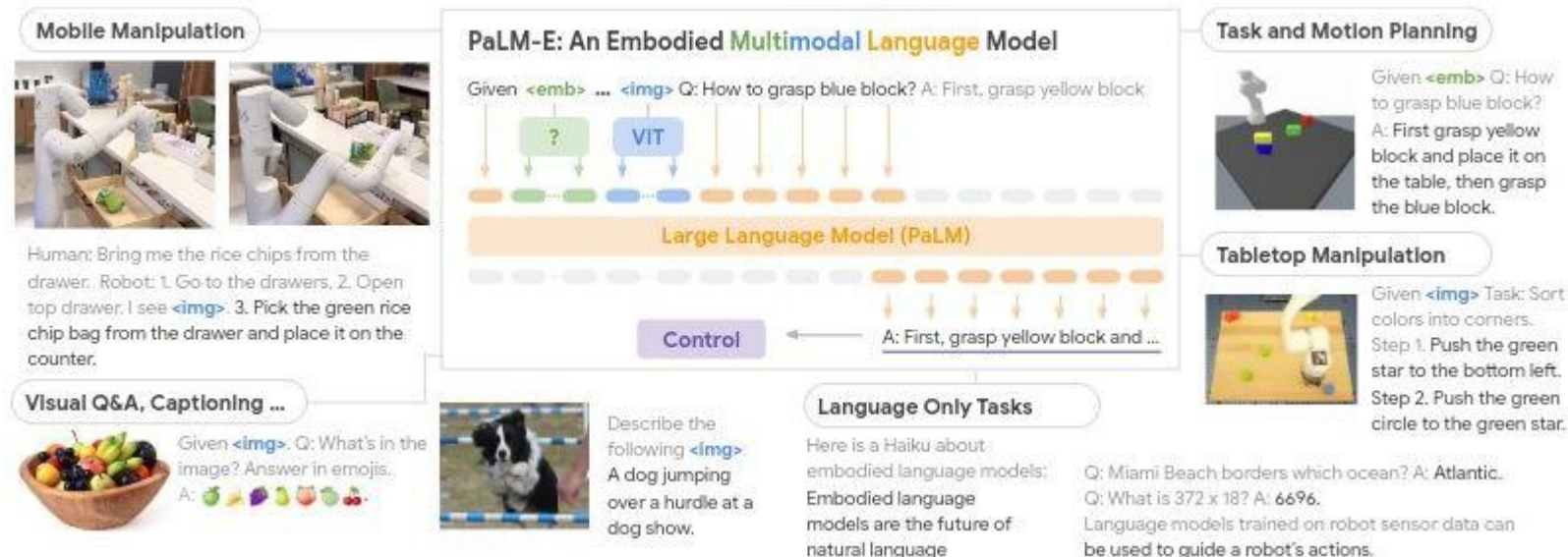
False



human : Bring me a drink from the table
Robot : Go to the table
Scene : I see a coke and lime soda
Robot : What kind of drink would you like ?
Human : Something with caffeine
Robot : Pick up coke
Success Detector : False

Examples of Models

Palm-E : An Embodied Multimodal Language Model



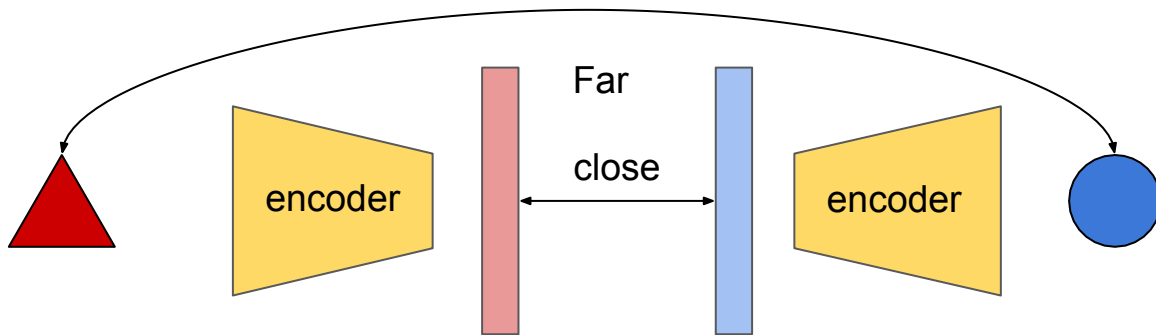
idea : Use the LLM to generate text to answer the task (VQA, Captioning, ...) and as a high-level policy that sequences and controls the low-level policies

Conclusion

- Use good architecture to encode your data

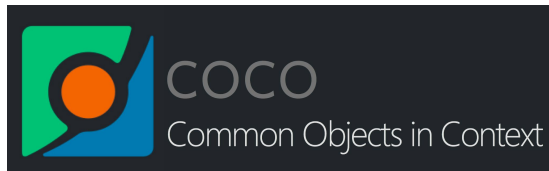
images	CNN
text	LSTM
video	LSTM / CNN
vectors	MLP

- Encode your data allow you to end up with closer representation of very different modalities



Conclusion

- Learn the task from the multimodal latent space is easier : Captioning, VQA, ...
- You can use pre-existing dataset or create your own to train your model



- Nowadays the most popular architecture is the Transformer : can handle multiple modality and is very powerful to learn links between tokens
 - BUT : hard to train → lot of data
- A good practice is also to use foundation models to turn the knowledge they acquired during training to good account